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# FORGE: REACTION-GUIDED GENERATIVE DESIGN OF IONIZABLE LIPIDS

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## ABSTRACT

Ionizable lipids are essential for RNA delivery, but their discovery largely depends on synthesizing and screening libraries assembled from predefined reactants. This strategy preserves a clear synthesis route while limiting the chemical space that can be explored. De novo molecular generation could expand this space, yet existing whole-molecule models do not retain the reaction information needed to assemble their designs. We introduce **FORGE**(**F**low-Matched, **O**pen-ended **R**eaction-**G**uided **G**eneration **E**xploration), a synthesis-grounded generative framework that uses known reaction chemistry to design ionizable lipids without restricting generation to a fixed reactant catalogue. FORGE represents each synthesis route through precursor roles, reaction-core positions, transformation order and coarse molecular architecture, while generating the complete molecule at atom-and-bond resolution. The model does not receive component identifiers, stored precursor graphs or fragment tokens. We evaluated FORGE on an AGILE-type Ugi three-component reaction, repeated aza-Michael addition and repeated reductive amination. Under population-optimal primary denoising and exact continuous-time integration, we prove that the ideal terminal state law recovers the encoded program-compatible endpoint distribution and therefore decodes to valid, verified exact-L1 graphs almost surely. Across three independent seeds, FORGE achieved verified exact level-one (exact-L1) transform-consistent yields of  $78.7 \pm 10.1\%$ ,  $77.5 \pm 3.7\%$  and  $31.3 \pm 14.3\%$ , respectively. Matched whole-molecule generation followed by post-hoc reaction filtering yielded  $11.4 \pm 1.1\%$ ,  $4.0 \pm 0.8\%$  and  $13.2 \pm 1.3\%$ ; the paired differences were 67.24, 73.49 and 18.10 percentage points. FORGE also generated products with verifier-recovered exact decompositions containing precursor constitutions absent from the corresponding training catalogues in all three reactions. In a bounded, route-blinded Ugi shortlist, 19 of 255 admitted products closed to complete computational synthesis dossiers; the remaining products were explicitly unresolved or search-censored rather than called unsynthesizable.

## 1 INTRODUCTION

Lipid nanoparticles (LNPs) deliver nucleic-acid therapeutics including small interfering RNA, mRNA vaccines and in vivo genome-editing reagents (Adams et al., 2018; Polack et al., 2020; Baden et al., 2021; Gillmore et al., 2021). Within each LNP, the ionizable lipid supports nucleic-acid complexation, particle assembly and endosomal escape, making its chemical structure a major determinant of delivery (Cullis & Hope, 2017; Hou et al., 2021). Small changes to the headgroup or hydrophobic domains can substantially alter function. Their effects depend on the complete molecular structure and do not follow simple design rules (Cheng et al., 2020; Han et al., 2021; Hatit et al., 2023). Researchers therefore discover ionizable lipids by defining modular precursor libraries, synthesizing combinations and screening the resulting LNPs experimentally (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). This strategy preserves synthetic tractability and supports systematic testing within defined chemical families, while limiting the search to components selected before design begins.

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De novo molecular generation has the potential to extend ionizable-lipid discovery beyond fixed precursor libraries. Current synthesis-aware generators construct molecules by selecting reactants and applying predefined transformations (Gao et al., 2022; Koziarski et al., 2024; Cretu et al., 2025). This construction retains an explicit assembly route and limits generated molecules to the supplied reactant inventory. Whole-molecule generators can propose structures beyond a fixed inventory, but their outputs do not retain the precursor roles and transformations needed to assemble each design (Qin et al., 2025; Lee et al., 2025). Other approaches estimate synthesizability after generation without incorporating the intended assembly chemistry into the generative process (Gao & Coley, 2020); the filter can only accept or reject a finished molecule. Recent studies have also applied de novo generative models to ionizable lipids (Ou et al., 2024). The missing capability is generation of complete molecular structures under known reaction chemistry while leaving precursor identities open.

We introduce FORGE, which conditions generation on a reaction program that specifies the assembly chemistry, precursor roles, reaction-core positions, transformation order and coarse molecular architecture, while leaving precursor identities unspecified. FORGE generates complete molecular graphs at atom-and-bond resolution without component identifiers, stored precursor graphs or fragment tokens. Each completed graph is decomposed into its implied precursors, and exact forward replay of the specified assembly reaction must reconstruct the generated molecule. For precursors that are not directly available, a bounded recursive search returns either a computational synthesis dossier supported by reaction and supplier evidence or an explicit unresolved or search-censored outcome.

We evaluate one shared model across an AGILE-type amine-aldehyde-isocyanide Ugi three-component reaction, repeated aza-Michael addition and repeated reductive amination. The Ugi reaction joins an amine head, an aldehyde and an isocyanide in a single step into an  $\alpha$ -amino amide carrying two hydrophobic regions (Xu et al., 2024). The other two attach hydrophobic components one at a time to the reactive amines of a single head, by conjugate addition of an amine to an acrylate (Li et al., 2023) or by reductive amination of an aldehyde (Xue et al., 2024). Each hydrophobic component carries one reactive handle, so the head’s reactive sites bound how many attach. These experiments test shared learning across three specified reaction programs, not zero-shot support for arbitrary chemistry. This short paper reports the principal computational findings; Appendix A through Appendix F give the complete formulation, the full reporting contract, the theoretical results and a sample of generated structures. We summarize our main contributions as follows.

- **Open-ended whole-graph generation.** FORGE generates complete molecular graphs using known reaction chemistry while leaving precursor identities open, and does not select from stored reactants or fragment vocabularies.
- **Conditioning acts on generation, not on filtering.** Matched controls across three assembly programs show that reaction-program conditioning increases exact transform-consistent generation relative to whole-molecule generation followed by post-hoc filtering.
- **Verified decompositions with component-novel precursors.** Across all three programs, FORGE generates verified exact assemblies with recovered precursor constitutions absent from the corresponding training component catalogues. Strong product-level catalogue escape is reserved for a direct common-reference reachability test and is not claimed here.
- **Ideal population correctness.** We prove that a population minimizer of the primary coordinatewise denoising loss recovers the local generator of the encoded endpoint path and, under exact continuous-time simulation, transports the program-compatible source to the conditional endpoint law. Its decoder pushforward is consequently valid and verified exact-L1 almost surely. The result deliberately excludes auxiliary losses, PCGrad, finite capacity and numerical integration error.

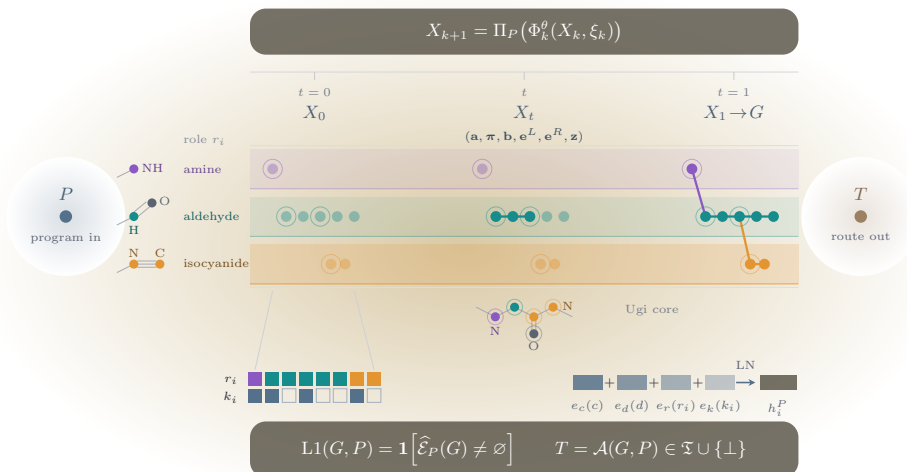


Figure 1: **FORGE generates complete ionizable-lipid graphs within reaction-program coordinates.** A reaction program specifies how molecular regions participate in an assembly without choosing their structures from a finite inventory. FORGE generates the whole graph, verifies that the specified reaction reconstructs it, then recursively expands unavailable precursors, terminating each branch in an evidence-qualified starting material or an explicit abstention.  $\Pi_P$  restores the coordinates the program fixes and leaves the rest as sampled; haloed positions are the reaction-core positions  $k_i$  marks. In the Ugi-core inset each atom takes the colour of its source component, and the carbonyl oxygen is grey because the template introduces it. Replay and route closure are computational checks, not evidence of experimental synthesis.

## 2 FORGE

FORGE generates complete ionizable-lipid graphs under specified assembly chemistry while leaving precursor identities open. Let  $G \in \mathcal{G}$  denote a complete ionizable-lipid graph and  $P \in \mathcal{P}$  a reaction program. FORGE models  $p_\theta(G \mid P)$  without access to precursor identifiers, stored component graphs or fragment tokens. Given  $P$ , FORGE must generate  $G$  and either construct a complete synthesis dossier or return a typed failure. Writing  $\mathfrak{D}$  for the dossier space and  $\mathfrak{F}_{\mathfrak{S}}$  for the finite failure set under the frozen route contract,

$$G \sim p_\theta(\cdot \mid P), \quad Y = \text{Assess}_{\mathfrak{S}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{S}}, \quad (1)$$

where  $(\mathbf{C}_T, \tau_T)$  is the sole verifier-returned exact root trace; ambiguous roots cause an abstention. The bounded constructor  $\text{Assess}_{\mathfrak{S}}$  verifies the final assembly, searches for preparation routes to unavailable precursors, and checks terminal supplier evidence.  $P$  guides generation, whereas the assessor constructs  $T$  from the completed graph without modifying it. FORGE accepts  $(G, T)$  only when  $G$  passes molecular and exact-assembly verification and  $T$  traces every required precursor to an available starting material; otherwise it abstains.

Figure 1 summarizes how the reaction program enters generation, what the flow is free to resample, and what the assembly reaction forms.

**Reaction programs.** A reaction program is a tuple  $P = (c, d, \mathbf{r}, \mathbf{k}, M)$ , where  $c$  identifies the ordered assembly chemistry,  $d$  is the program depth,  $\mathbf{r}$  assigns precursor roles,  $\mathbf{k}$  marks reaction-core positions, and  $M$  specifies coarse per-role morphology as four integers per role: exterior atom count, junction budget, cycle rank and attachment count.  $M$  constrains coarse molecular architecture while leaving the exact topology, atoms, bonds and precursor identities unspecified. FORGE samples an admissible complete program  $P$  from a training-fold prior and then generates  $G$  conditioned on it, so that  $p_\theta(G, P \mid c) = p_{\text{prog}}(P \mid c) p_\theta(G \mid P)$ . At graph position  $i$ , the program embedding combines learned embeddings of program iden-

tity, program depth, precursor role, reaction-core position and the four role-local morphology counts. Those counts also define active positions and masks. The embedding identifies the position’s semantic layout without encoding a component identity.

**Discrete flow generation.** FORGE proceeds in four stages: the program specifies roles, core positions, order and coarse morphology; a conditional discrete flow generates every atom, bond and topology state under it; a reverse transform decomposes the product and the corresponding forward transform must reconstruct it exactly; and precursors that are not already available are expanded recursively or the design is withheld. Generation operates on a sparse tree-plus-closure encoding. Given the selected exact atom-mapped annotation  $\alpha$ , let  $X_G := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z})$  be the categorical graph state, with  $\mathbf{a}$  atom states,  $\boldsymbol{\pi}$  parent pointers,  $\mathbf{b}$  spanning-tree bond states,  $\mathbf{e}^L$  and  $\mathbf{e}^R$  closure endpoints and  $\mathbf{z}$  closure-bond states. The decoder is surjective onto the declared graph space  $\mathfrak{G}_{N_{\max}, K_{\text{cl}}}$ , here  $N_{\max} = 194$  and  $K_{\text{cl}} = 3$ , so every graph within the declared atom, bond and size support is representable. This is a statement about architectural support, not about reaction consistency or probability under a trained model.

The chemistry adapter marks the atom, pointer and bond states that  $P$  fixes. FORGE holds those constant while corrupting the remaining states along the linear probability path and predicting their clean values from  $(X_t, t, P)$ . Writing  $\mathcal{J}$  for the six state families,  $m_j^{(q)}(P) \in \{0, 1\}$  for whether coordinate  $j$  remains variable under  $P$ ,  $\mu_{\text{bal}}$  for the frozen program-balanced training distribution, and  $\ell_j^{(q)}(\theta) = -\log p_\theta(X_{G,j}^{(q)} \mid X_t, t, P)$  for the per-coordinate denoising term, FORGE minimizes

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G,P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0,1), X_t \sim p_t(\cdot \mid X_G, P)} \left[ \sum_{q \in \mathcal{J}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \ell_j^{(q)}(\theta)}{\max\{1, \sum_j m_j^{(q)}(P)\}} \right], \quad (2)$$

where  $\lambda_q$  weights family  $q$  and a family with no variable coordinates contributes zero loss. The corruption and sampling operators give fixed coordinates zero update rate and project each finite-step state back onto them. This is the precise syntactic sense in which the program constrains the trajectory: fixed coordinates cannot be resampled. Chemical consistency is established separately by exact forward replay. The denoising result in Appendix E concerns the primary masked term only, not auxiliary losses or PCGrad. Appendix B gives the sampler, verifier and bounded route assessor in full, and Appendix E the representation, invariance, sampling-efficiency and termination results.

**Ideal population guarantee.** The following result closes the idealized chain from encoded endpoint data, through the primary denoising posterior and local continuous-time generator, to the decoded terminal graph law. Write  $\widehat{\text{L1}}(G, P) = 1$  when the frozen verifier returns at least one program-compatible decomposition that exactly reconstructs  $G$  by forward replay.

**Theorem 2.1** (Ideal population correctness of program-conditioned generation). *Fix a program  $P$  with positive mass under  $\mu_{\text{bal}}$ , and let*

$$q_1^P = \mathcal{L}(\text{Enc}_P(G, \alpha) \mid P), \quad \mu_G^P = \mathcal{L}([G]_{\cong} \mid P) \quad (3)$$

*be the encoded endpoint law and corresponding conditional graph law. Assume finite coordinate alphabets and a full-support variable-coordinate source; every admitted endpoint decodes by exact round trip, is molecularly valid and satisfies  $\widehat{\text{L1}}(G, P) = 1$ ; the unrestricted coordinatewise model is at a population minimum of the primary masked denoising term; and sampling starts from the program source and exactly simulates the induced local continuous-time Markov chain. Then, writing  $G_1 = \text{Dec}(X_1)$ , the ideal terminal state satisfies*

$$\mathcal{L}(X_1 \mid P) = q_1^P, \quad \text{Dec}_{\#} q_1^P = \mu_G^P, \quad (4)$$

$$\Pr[\text{Valid}(G_1) = 1, \widehat{\text{L1}}(G_1, P) = 1 \mid P] = 1. \quad (5)$$

Appendix E defines the multivariate probability path and local generator and proves Theorem 2.1. The result is deliberately ideal: it does not characterize the trained finite-capacity Transformer, auxiliary objectives, PCGrad update or capped finite-step sampler. It also neither implies strong catalogue escape nor assigns positive probability to graphs outside the endpoint law.

### 3 RESULTS

**Setting.** We evaluated three assembly programs, with Ugi as the deeply supervised case, repeated aza-Michael addition as a second family, and repeated reductive amination as a lighter stress test, across four matched training arms: Ugi-only conditioning, shared conditioning, shared training without program information, and shared training with cyclically incorrect program labels. All arms shared architecture, optimization budget and attempts per program, and we evaluated the prespecified final checkpoint from each of three independent seeds. The primary metric is verified exact-L1 yield per generation attempt: generated graphs passing molecular validity, program-aware decomposition and exact forward replay, divided by total attempts. Training used only the component-disjoint training folds, and held-out products neither entered training nor selected checkpoints. Generated molecules are not treated as independent replicates. Appendix C states the full protocol, including the external-baseline admission rules.

**Generalizing across reaction programs.** Across 3,072 held-out generation attempts per program in each of three independent training seeds, the conditioned model achieved verified exact-L1 yields of  $78.7 \pm 10.1\%$  for Ugi (72.3%, 90.3% and 73.5% by seed),  $77.5 \pm 3.7\%$  for repeated aza-Michael addition (76.5%, 81.5% and 74.3%), and  $31.3 \pm 14.3\%$  for repeated reductive amination (38.6%, 14.8% and 40.4%) (mean  $\pm$  sample standard deviation; Table 1). Every verifier-returned trace passed forward replay, as required by the verifier invariant. Mean exact-decomposition ambiguity was 54.9% for aza-Michael addition and 74.1% for reductive amination, and reductive amination also varied substantially between seeds, indicating less consistent learning in this lower-data setting. Relative to the Ugi-only model, shared training changed verified Ugi exact-L1 yield by 53.26 percentage points; the paired seed differences were 41.24, 56.71, 61.82 points, spanning 41.24 to 61.82. The observed runs met the numerical Ugi-retention gate. As an implementation check, they recorded zero fixed-coordinate violations and zero support overflows across 27,648 held-out attempts. One conditional flow learned the three specified reaction-program coordinates without erasing the deeply supported Ugi behavior.

Table 1: **Shared learning across three specified assembly programs.** Verified exact-L1 yield (%) and transform verification per program, as mean  $\pm$  sample standard deviation over three training seeds. Results are never pooled across programs. Per-arm and per-seed values: Appendix C. Decomposition coverage is conditioned on valid products. Replay is a verifier invariant, not a measure of learned performance.

| Program             | FORGE conditioned | Shared null    | Cyclic program | Decomp. coverage /<br>replay invariant (%) |
|---------------------|-------------------|----------------|----------------|--------------------------------------------|
| Ugi                 | $78.7 \pm 10.1$   | $11.4 \pm 1.1$ | $40.2 \pm 0.7$ | 90.7/100.0                                 |
| Aza-Michael         | $77.5 \pm 3.7$    | $4.0 \pm 0.8$  | $3.8 \pm 0.6$  | 100.0/100.0                                |
| Reductive amination | $31.3 \pm 14.3$   | $13.2 \pm 1.3$ | $18.1 \pm 0.8$ | 84.3/100.0                                 |

**Reaction information changes generation rather than merely filtering its output.** Under matched attempt budgets, conditioned generation produced 786.8, 774.6 and 312.7 exact-L1 products per 1,000 attempts for Ugi, repeated aza-Michael and repeated reductive amination, compared with 114.4, 39.7 and 131.7 for matched null generation followed by post-hoc filtering. The mean paired conditioned-minus-post-hoc differences were 67.24 percentage points for Ugi (three-seed range 60.84 to 79.95), 73.49 for aza-Michael (69.69 to 78.45), and 18.10 for reductive amination (2.51 to 26.04). A cyclic control that retained

precursor roles, reaction-core coordinates and program depth but assigned the wrong reaction identity changed verified exact-L1 yield by 38.43 percentage points for Ugi (three-seed range 32.00 to 49.41), 73.67 for aza-Michael and 13.15 for reductive amination. Changing only program coordinates at inference, with the trained network and Ugi request fixed, reduced verified exact-L1 yield from 39.69% to 3.62% when all coordinates were nulled (36.07 points; paired differences 37.24, 34.28, 36.69, spanning 34.28 to 37.24), whereas the aza-Michael-token contrast was unresolved at 2.00 points. The reaction program therefore changes molecular generation before downstream verification, but the evidence does not support an equal or individually identifiable effect for every coordinate.

**Whole-graph generation recovers precursor constitutions outside training catalogues.** Table 2 compares FORGE with external synthesis-aware generators, matched controls and a finite-catalogue oracle under one common Ugi benchmark, normalizing every yield by generation attempts. The learned finite-inventory selector reached  $1000.0 \pm 0.0$  verified exact-L1 products per 1,000 attempts with zero verifier-recovered component-novel yield, imposed by its finite action space, whereas FORGE produced  $786.8 \pm 100.8$  verified exact-L1 and  $751.2 \pm 80.2$  distinct verified exact-L1 products. Among whole-molecule external generators, unconditional DeFoG produced  $280.6 \pm 26.6$  verified exact-L1 products and recovered held components at  $15.5 \pm 2.4$  per 1,000 attempts against  $0.1 \pm 0.2$  for FORGE; GenMol/SAFE produced  $627.5 \pm 6.2$  valid connected outputs but no exact Ugi-L1 product, and RGFN emitted no admissible output under the strict native contract. Against the train-only assembler, FORGE effective component counts were 320.8, 156.9 and 102.9 versus 86.4, 23.4 and 16.9. Reaction consistency, verifier-recovered component novelty and recovery of reserved components therefore trade off separately. This component-level diagnostic is not a direct test of product-level strong catalogue escape. Representative Ugi products show exact decompositions into precursor identities outside the visible training inventory (Figure 2).

Table 2: **Common Ugi benchmark for external generators and matched controls.** Yields per 1,000 generation attempts under one component-disjoint split, three-seed protocol and common exact-L1 verifier (mean  $\pm$  sample standard deviation). Component-novel: exactly one verifier-returned decomposition with a component constitution absent from the method-visible training catalogue. This verifier-relative metric is not strong catalogue escape. The whole-molecule baselines have no reaction program, so their exact-L1 yields come from the same post-hoc assessment applied to every method.  $\dagger$ , a zero imposed by a finite component action space, not an empirical failure; N/E, not estimable. No published values from the original baseline papers are inserted.  $\uparrow$  marks a column in which higher is better. Diversity carries no arrow because it is diagnostic rather than directional: high mean pairwise distance over a collapsed effective component count is not a better result, and is read against  $N_{\text{eff}}$  in Table 4. Definitions and seed-level values: Appendix C.

| Models                     | Valid $\uparrow$  | Verified L1 $\uparrow$ | Distinct L1 $\uparrow$ | Comp.-novel $\uparrow$ | Held-comp. $\uparrow$ | Diversity         |
|----------------------------|-------------------|------------------------|------------------------|------------------------|-----------------------|-------------------|
| RGFN                       | $0.0 \pm 0.0$     | $0.0 \pm 0.0$          | $0.0 \pm 0.0$          | $0.0 \pm 0.0^\dagger$  | $0.0 \pm 0.0$         | N/E               |
| DeFoG unconditional        | $346.5 \pm 42.3$  | $280.6 \pm 26.6$       | $280.6 \pm 26.6$       | $280.6 \pm 26.6$       | $15.5 \pm 2.4$        | $0.727 \pm 0.014$ |
| GenMol/SAFE                | $627.5 \pm 6.2$   | $0.0 \pm 0.0$          | $0.0 \pm 0.0$          | $0.0 \pm 0.0$          | $0.0 \pm 0.0$         | $0.873 \pm 0.009$ |
| Learned inventory selector | $1000.0 \pm 0.0$  | $1000.0 \pm 0.0$       | $849.5 \pm 13.7$       | $0.0 \pm 0.0^\dagger$  | $0.0 \pm 0.0$         | $0.717 \pm 0.001$ |
| Finite catalogue oracle    | $1000.0 \pm 0.0$  | $1000.0 \pm 0.0$       | $837.6 \pm 8.2$        | $0.0 \pm 0.0^\dagger$  | $0.0 \pm 0.0$         | $0.719 \pm 0.000$ |
| Shared-null + post-hoc     | $194.7 \pm 15.6$  | $114.4 \pm 10.9$       | $108.5 \pm 9.7$        | $108.5 \pm 9.7$        | $0.4 \pm 0.2$         | $0.754 \pm 0.014$ |
| FACT-matched               | $60.8 \pm 20.2$   | $52.0 \pm 15.6$        | $50.8 \pm 14.4$        | $50.8 \pm 14.4$        | $0.0 \pm 0.0$         | $0.667 \pm 0.009$ |
| FACT-generous              | $500.7 \pm 213.8$ | $296.7 \pm 186.0$      | $288.3 \pm 175.3$      | $288.3 \pm 175.3$      | $0.1 \pm 0.2$         | $0.774 \pm 0.007$ |
| FORGE Transformer          | $866.0 \pm 90.2$  | $786.8 \pm 100.8$      | $751.2 \pm 80.2$       | $751.2 \pm 80.2$       | $0.1 \pm 0.2$         | $0.621 \pm 0.011$ |

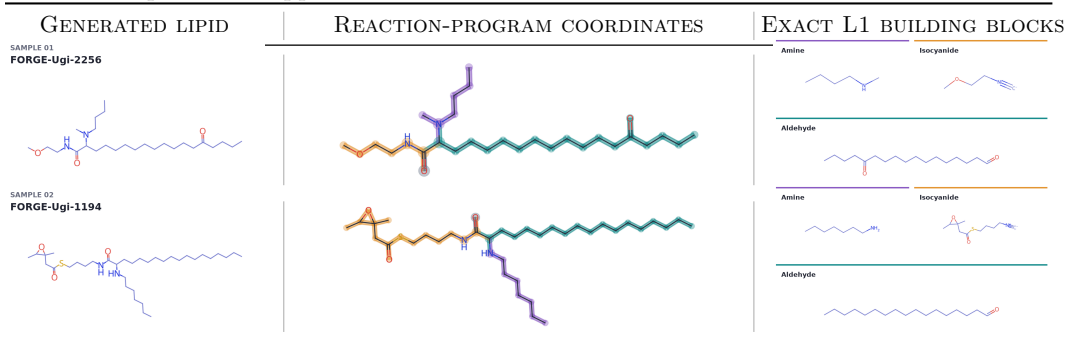
**Recursive synthesis programs connect generated lipids to evidence-qualified materials.** Verified exact-L1 establishes the specified final assembly reaction but does not establish a complete synthesis route. Route assessment is method-blind, and a reaction sequence counts as a complete dossier only when the final assembly and every upstream step pass exact forward verification and every terminal material closes under the frozen evidence policy. The frozen union contained 11,313 distinct role-constitution components, of which

the exact-source library closed 29; FORGE therefore produced  $786.8 \pm 100.8$  verified exact-L1 products per 1,000 attempts but  $0.0 \pm 0.0$  complete dossiers and  $786.8 \pm 100.8$  abstentions under that library. In the bounded cascade over a frozen, route-blinded shortlist of 256 distinct Ugi products, all 256 passed exact-L1 reverification, 255 entered route assessment, and 19 closed, with 236 unresolved and 1 search-censored. Most unresolved components were absent from the bounded evidence index and were never expanded, so this measures closure under a frozen cascade rather than molecular synthesizability. We record incomplete bounded route searches as abstentions because they do not demonstrate unsynthesizability (Table 3).

Table 3: **Bounded route assessment returns auditable outcomes.** The route-blinded shortlist contains distinct Ugi products that passed exact-L1 reverification. Complete denotes closure under the frozen upstream-reaction evidence and terminal-material policy; unresolved and censored outcomes are abstentions, not evidence of unsynthesizability.

| Unit              | Population | Complete | Unresolved | Censored |
|-------------------|------------|----------|------------|----------|
| Products          | 256        | 19       | 236        | 1        |
| Unique components | 158        | 36       | 120        | 2        |

Figure 2: **Deterministic examples of component-novel FORGE ionizable-lipid graphs.** Each row is one seed-0 Ugi product with a unique exact-L1 decomposition and a recovered precursor constitution outside the method-visible training inventory. Left, the graph sampled jointly by FORGE; centre, precursor origins recovered by exact atom-mapped forward replay, halos marking the Ugi core; right, the exact precursor constitutions returned by the verifier. Purple denotes the amine-derived head, teal the aldehyde-derived body and tail, orange the isocyanide-derived tail and grey the assembly-introduced oxygen; these are semantic coordinates recovered by replay, not a vocabulary the model selects from. These are display examples, not experimental candidates, and exact-L1 replay is a structural diagnostic, not evidence of activity or synthesis success. Selection rule and the complete twelve-example atlas: Appendix F.



## 4 DISCUSSION

We present FORGE, a generative framework that uses known assembly chemistry during whole-molecule generation while leaving precursor identities open. Across an AGILE-type Ugi three-component reaction, repeated aza-Michael addition and repeated reductive amination, reaction-program conditioning increased verified exact-L1 yield relative to matched whole-molecule generation followed by post-hoc filtering, and FORGE generated exact assemblies whose verifier-returned traces contained precursor identities absent from the corresponding training catalogues. Reaction information can therefore guide molecular generation without making a finite component inventory part of the generative state. Because direct catalogue-product reachability was not computed, this study does not claim strong product-level catalogue escape.

The evidence is strongest for AGILE-type Ugi 3-CR; repeated aza-Michael addition is a second computational family and repeated reductive amination a lighter stress test, showing lower exact-L1 yield, greater decomposition ambiguity and substantial between-seed variation. The study is computational, supports only three qualified assembly programs and operates within a bounded molecular graph space. Exact forward replay verifies structural consistency with the final assembly reaction but does not establish precursor availability, experimental synthesis success, lipid activity or RNA delivery, and the bounded HeLa diagnostic does not establish biological activity. Several prespecified outcomes did not favor FORGE and are reported in full in Appendix C: native recovery of specifically reserved held components remained weak at 0.11 products per 1,000 attempts, no evaluated method reproduced the held-out observed-lipid distribution, joint denoising was not uniformly superior to role isolation across programs, and neither synthesis tilting nor the nested potency proposal improved its matched terminal endpoint. Prospective synthesis, formulation and biological validation are outside the scope of this computational paper; none of the reported metrics should be interpreted as such evidence. Within these limits, reaction programs provide a practical coordinate system for expanding molecular identity without disconnecting generative design from assembly chemistry.

## REFERENCES

- David Adams, Alejandra Gonzalez-Duarte, William D. O’Riordan, Chih-Chao Yang, Mitsuharu Ueda, Arnt V. Kristen, Ivailo Tournev, Hartmut H. Schmidt, Teresa Coelho, John L. Berk, Kon-Ping Lin, Giuseppe Vita, Shahram Attarian, Violaine Planté-Bordeneuve, Michelle M. Mezei, Josep M. Campistol, Juan Buades, Thomas H. Brannagan, Byoung J. Kim, Jeeyoung Oh, Yesim Parman, Yoshiki Sekijima, Philip N. Hawkins, Scott D. Solomon, Michael Polydefkis, Peter J. Dyck, Pritesh J. Gandhi, Sunita Goyal, Jihong Chen, Andrew L. Strahs, Saraswathy V. Nochur, Marianne T. Sweetser, Pushkal P. Garg, Akshay K. Vaishnaw, Jared A. Gollob, and Ole B. Suhr. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *New England Journal of Medicine*, 379(1):11–21, 2018. doi: 10.1056/NEJMoa1716153.
- Akin Akinc, Andreas Zumbuehl, Michael Goldberg, Elizaveta S. Leshchiner, Valentina Busini, Naushad Hossain, Sergio A. Bacallado, David N. Nguyen, Jason Fuller, Rene Alvarez, Anna Borodovsky, Todd Borland, Rainer Constien, Antonin de Fougerolles, J. Robert Dorkin, K. Narayanannair Jayaprakash, Muthusamy Jayaraman, Matthias John, Victor Kotliansky, Muthiah Manoharan, Lubomir Nechev, June Qin, Timothy Racie, Denitza Raitcheva, Kallanthottathil G. Rajeev, Dinah W. Y. Sah, Jürgen Soutschek, Ivanka Toudjarska, Hans-Peter Vornlocher, Tracy S. Zimmermann, Robert Langer, and Daniel G. Anderson. A combinatorial library of lipid-like materials for delivery of RNAi therapeutics. *Nature Biotechnology*, 26(5):561–569, 2008. doi: 10.1038/nbt1402.
- Lindsey R. Baden, Hana M. El Sahly, Brandon Essink, Karen Kotloff, Sharon Frey, Rick Novak, David Diemert, Stephen A. Spector, Nadine Rouphael, C. Buddy Creech, John McGettigan, Shishir Khetan, Nathan Segall, Joel Solis, Adam Brosz, Carlos Fierro, Howard Schwartz, Kathleen Neuzil, Lawrence Corey, Peter Gilbert, Holly Janes, Dean Follmann, Mary Marovich, John Mascola, Laura Polakowski, Julie Ledgerwood, Barney S. Graham, Hamilton Bennett, Rolando Pajon, Conor Knightly, Brett Leav, Weiping Deng, Honghong Zhou, Shu Han, Melanie Ivarsson, Jacqueline Miller, and Tal Zaks. Efficacy and safety of the mRNA-1273 SARS-CoV-2 vaccine. *New England Journal of Medicine*, 384(5):403–416, 2021. doi: 10.1056/NEJMoa2035389.
- Qiang Cheng, Tuo Wei, Lukas Farbiak, Lindsay T. Johnson, Sean A. Dilliard, and Daniel J. Siegwart. Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR-Cas gene editing. *Nature Nanotechnology*, 15(4):313–320, 2020. doi: 10.1038/s41565-020-0669-6.
- Miruna Cretu, Charles Harris, Ilia Igashov, Arne Schneuing, Marwin Segler, Bruno Correia, Julien Roy, Emmanuel Bengio, and Pietro Liò. SynFlowNet: Design of diverse and novel

- 
- molecules with synthesis constraints. In *International Conference on Learning Representations*, 2025.
- Pieter R. Cullis and Michael J. Hope. Lipid nanoparticle systems for enabling gene therapies. *Molecular Therapy*, 25(7):1467–1475, 2017. doi: 10.1016/j.ymthe.2017.03.013.
- Wenhao Gao and Connor W. Coley. The synthesizability of molecules proposed by generative models. *Journal of Chemical Information and Modeling*, 60(12):5714–5723, 2020. doi: 10.1021/acs.jcim.0c00174.
- Wenhao Gao, Rocío Mercado, and Connor W. Coley. Amortized tree generation for bottom-up synthesis planning and synthesizable molecular design. In *International Conference on Learning Representations*, 2022.
- Samuel Genheden, Amol Thakkar, Veronika Chadimová, Jean-Louis Reymond, Ola Engkvist, and Esben Bjerrum. AiZynthFinder: A fast, robust and flexible open-source software for retrosynthetic planning. *Journal of Cheminformatics*, 12:70, 2020. doi: 10.1186/s13321-020-00472-1.
- Julian D. Gillmore, Ed Gane, Jorg Taubel, Justin Kao, Marianna Fontana, Michael L. Maitland, Jessica Seitzer, Daniel O’Connell, Kathryn R. Walsh, Kristy Wood, Jonathan Phillips, Yuanxin Xu, Adam Amaral, Adam P. Boyd, Jeffrey E. Cehelsky, Mark D. McKee, Andrew Schiermeier, Olivier Harari, Andrew Murphy, Christos A. Kyratsous, Brian Zambrowicz, Randy Soltys, David E. Gutstein, John Leonard, Laura Sepp-Lorenzino, and David Lebowitz. CRISPR-Cas9 in vivo gene editing for transthyretin amyloidosis. *New England Journal of Medicine*, 385(6):493–502, 2021. doi: 10.1056/NEJMoa2107454.
- Xuexiang Han, Hanwen Zhang, Kamila Butowska, Kelsey L. Swingle, Mohamad-Gabriel Alameh, Drew Weissman, and Michael J. Mitchell. An ionizable lipid toolbox for RNA delivery. *Nature Communications*, 12:7233, 2021. doi: 10.1038/s41467-021-27493-0.
- Marine Z. C. Hatit, Curtis N. Dobrowolski, Melissa P. Lokugamage, David Loughrey, Huanzhen Ni, Chiara Zurla, Alejandro J. Da Silva Sanchez, Afsane Radmand, Sebastian G. Huayameres, Ryan Zenhausern, Kalina Paunovska, Hannah E. Peck, Jinwhan Kim, Manaka Sato, Jacob I. Feldman, Michael-Alexander Rivera, Ana Cristian, Yong-Tae Kim, Philip J. Santangelo, and James E. Dahlman. Nanoparticle stereochemistry-dependent endocytic processing improves in vivo mRNA delivery. *Nature Chemistry*, 15(4):508–515, 2023. doi: 10.1038/s41557-023-01138-9.
- Xucheng Hou, Tal Zaks, Robert Langer, and Yizhou Dong. Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials*, 6(12):1078–1094, 2021. doi: 10.1038/s41578-021-00358-0.
- Muthusamy Jayaraman, Steven M. Ansell, Barbara L. Mui, Ying K. Tam, Jianxin Chen, Xinyao Du, David Butler, Laxman Eltepu, Shigeo Matsuda, Jayaprakash K. Narayanan-nair, Kallanthottathil G. Rajeev, Ismail M. Hafez, Akin Akinc, Martin A. Maier, Mark A. Tracy, Pieter R. Cullis, Thomas D. Madden, Muthiah Manoharan, and Michael J. Hope. Maximizing the potency of siRNA lipid nanoparticles for hepatic gene silencing in vivo. *Angewandte Chemie International Edition*, 51(34):8529–8533, 2012. doi: 10.1002/anie.201203263.
- Michał Koziarski, Andrei Reakesh, Dmytro Shevchuk, Almer van der Sloot, Piotr Gaiński, Yoshua Bengio, Cheng-Hao Liu, Mike Tyers, and Robert Batey. RGFN: Synthesizable molecular generation using GFlowNets. In *Advances in Neural Information Processing Systems*, volume 37, pp. 46908–46955, 2024.
- Seul Lee, Karsten Kreis, Srimukh Prasad Veccham, Meng Liu, Danny Reidenbach, Yuxing Peng, Saeed Gopal Paliwal, Weili Nie, and Arash Vahdat. GenMol: A drug discovery generalist with discrete diffusion. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 33205–33226. PMLR, 2025.

- 
- Bowen Li, Rajith Singh Manan, Shun-Qing Liang, Akiva Gordon, Allen Jiang, Andrew Varley, Guangping Gao, Robert Langer, Wen Xue, and Daniel Anderson. Combinatorial design of nanoparticles for pulmonary mRNA delivery and genome editing. *Nature Biotechnology*, 41(10):1410–1415, 2023. doi: 10.1038/s41587-023-01679-x.
- Bowen Li, Idris O. Raji, Akiva G. R. Gordon, Lizhuang Sun, Theresa M. Raimondo, Favour A. Oladimeji, Allen Y. Jiang, Andrew Varley, Robert S. Langer, and Daniel G. Anderson. Accelerating ionizable lipid discovery for mRNA delivery using machine learning and combinatorial chemistry. *Nature Materials*, 23(7):1002–1008, 2024. doi: 10.1038/s41563-024-01867-3.
- Kevin T. Love, Kerry P. Mahon, Christopher G. Levins, Kathryn A. Whitehead, William Querbes, J. Robert Dorkin, June Qin, William Cantley, Liu Liang Qin, Timothy Racie, Maria Frank-Kamenetsky, Ka Ning Yip, Rene Alvarez, Dinah W. Y. Sah, Antonin de Fougères, Kevin Fitzgerald, Victor Kotliansky, Akin Akinc, Robert Langer, and Daniel G. Anderson. Lipid-like materials for low-dose, in vivo gene silencing. *Proceedings of the National Academy of Sciences*, 107(5):1864–1869, 2010. doi: 10.1073/pnas.0910603106.
- Yuxuan Ou, Jingyi Zhao, Austin Tripp, Morteza Rasoulboroujeni, and José Miguel Hernández-Lobato. A deep generative model for the design of synthesizable ionizable lipids. In *NeurIPS 2024 Workshop on AI for New Drug Modalities*, 2024. arXiv:2412.00928.
- Fernando P. Polack, Stephen J. Thomas, Nicholas Kitchin, Judith Absalon, Alejandra Gurtman, Stephen Lockhart, John L. Perez, Gonzalo Pérez Marc, Edson D. Moreira, Cristiano Zerbini, Ruth Bailey, Kena A. Swanson, Satrajit Roychoudhury, Kenneth Koury, Ping Li, Warren V. Kalina, David Cooper, Robert W. Frenc, Laura L. Hammitt, Özlem Türeci, Haylene Nell, Axel Schaefer, Serhat Ünal, Dina B. Tresnan, Susan Mather, Philip R. Dormitzer, Uğur Şahin, Kathrin U. Jansen, and William C. Gruber. Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine. *New England Journal of Medicine*, 383(27):2603–2615, 2020. doi: 10.1056/NEJMoa2034577.
- Yiming Qin, Manuel Madeira, Dorina Thanou, and Pascal Frossard. DeFoG: Discrete flow matching for graph generation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 50269–50326. PMLR, 2025.
- Andrei Rekesh, Miruna Cretu, Dmytro Shevchuk, Vignesh Ram Somnath, Pietro Liò, Robert A. Batey, Mike Tyers, Michał Koziarski, and Cheng-Hao Liu. SynCoGen: Synthesizable 3d molecule generation via joint reaction and coordinate modeling. arXiv:2507.11818, 2025.
- Sean C. Semple, Akin Akinc, Jianxin Chen, Ammen P. Sandhu, Barbara L. Mui, Connie K. Cho, Dinah W. Y. Sah, Derrick Stebbing, Erin J. Crosley, Ed Yaworski, Ismail M. Hafez, J. Robert Dorkin, June Qin, Kieu Lam, Kallanthottathil G. Rajeev, Kim F. Wong, Lloyd B. Jeffs, Lubomir Nechev, Merete L. Eisenhardt, Muthusamy Jayaraman, Mikameh Kazem, Martin A. Maier, Masuna Srinivasulu, Michael J. Weinstein, Qingmin Chen, Rene Alvarez, Scott A. Barros, Soma De, Sandra K. Klimuk, Todd Borland, Verbena Kosovrasti, William L. Cantley, Ying K. Tam, Muthiah Manoharan, Marco A. Ciufolini, Mark A. Tracy, Antonin de Fougères, Ian MacLachlan, Pieter R. Cullis, Thomas D. Madden, and Michael J. Hope. Rational design of cationic lipids for siRNA delivery. *Nature Biotechnology*, 28(2):172–176, 2010. doi: 10.1038/nbt.1602.
- Yue Xu, Shihao Ma, Haotian Cui, Jingan Chen, Shufen Xu, Fanglin Gong, Alex Golubovic, Muye Zhou, Kevin Chang Wang, Andrew Varley, Rick Xing Ze Lu, Bo Wang, and Bowen Li. AGILE platform: A deep learning powered approach to accelerate LNP development for mRNA delivery. *Nature Communications*, 15:6305, 2024. doi: 10.1038/s41467-024-50619-z.

---

Lulu Xue, Alex G. Hamilton, Gan Zhao, Zebin Xiao, Rakan El-Mayta, Xuexiang Han, Ningqiang Gong, Xinhong Xiong, Junchao Xu, Christian G. Figueroa-Espada, Sarah J. Shepherd, Alvin J. Mukalel, Mohamad-Gabriel Alameh, Jiaxi Cui, Karin Wang, Andrew E. Vaughan, Drew Weissman, and Michael J. Mitchell. High-throughput barcoding of nanoparticles identifies cationic, degradable lipid-like materials for mRNA delivery to the lungs in female preclinical models. *Nature Communications*, 15(1):1884, 2024. doi: 10.1038/s41467-024-45422-9.

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## A PRELIMINARIES

Appendices A through F are supplementary and are excluded from the five-page short-paper limit, as are the references. They give the full formulation, the complete reporting contract, the theoretical results and a generated-structure atlas.

### A.1 IONIZABLE-LIPID STRUCTURE AND MODULAR ASSEMBLY

The structure of an ionizable lipid is typically organized into three functional regions: a protonatable headgroup, a linker, and one or more hydrophobic domains (Semple et al., 2010; Jayaraman et al., 2012). Together, these regions determine the ionization, stability, lipid packing, and membrane interactions of the molecule. Ionizable-lipid libraries are often synthesized by applying a common reaction to different combinations of precursor molecules (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). Each precursor contributes a defined portion of the final lipid and contains the reactive group required for the assembly. The reaction forms the bonds that connect these precursor-derived regions. As a result, the assembly reaction determines how the regions are connected, while the precursor structures determine the molecular identity of the resulting lipid.

### A.2 ASSEMBLY CHEMISTRIES STUDIED

We study three ionizable-lipid assembly chemistries: an AGILE-type amine–aldehyde–isocyanide Ugi three-component reaction, repeated aza-Michael addition, and repeated reductive amination. In the AGILE-type Ugi reaction, an amine head reacts with an aldehyde component and an isocyanide component to form an  $\alpha$ -amino amide that connects the headgroup to two hydrophobic regions (Xu et al., 2024). A second assembly strategy uses repeated aza-Michael additions to attach acrylate-bearing hydrophobic components sequentially to reactive amines in the headgroup (Li et al., 2023). Reductive amination sequentially couples aldehyde-bearing hydrophobic components to reactive amines in the headgroup through new carbon–nitrogen bonds (Xue et al., 2024). Figure 3 draws all three.

### A.3 REACTION TRANSFORMS AND EXACT ASSEMBLY

Each assembly chemistry defines a forward transformation that maps role-assigned precursor molecules to the product graphs permitted by that reaction. The corresponding reverse transformation decomposes a complete lipid into the precursor assignments that could have produced it. To verify a decomposition, we computationally replay the reaction on the recovered precursors. The product passes the level-one (L1) assembly check only if the reconstruction matches the original lipid exactly.

### A.4 DISCRETE MOLECULAR GRAPH SPACES

Let  $\Sigma_{\text{atom}}$  and  $\Sigma_{\text{bond}}$  be finite atom-state and bond-state vocabularies. A molecular graph is  $G = (V, E, \ell_V, \ell_E)$ , where the underlying graph is finite, connected, undirected and simple,  $\ell_V : V \rightarrow \Sigma_{\text{atom}}$ , and  $\ell_E : E \rightarrow \Sigma_{\text{bond}}$ . Bond order is part of  $\ell_E$  rather than a parallel edge. Graphs are identified up to label-preserving isomorphism. The declared model support additionally bounds the heavy-atom count and cycle rank.

### A.5 DISCRETE FLOW MATCHING

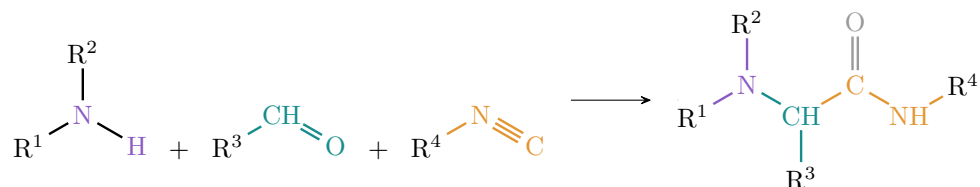
For one coordinate with alphabet  $\mathcal{X}_i$ , full-support source marginal  $q_{0,i}$  and clean endpoint  $a \in \mathcal{X}_i$ , the linear conditional path is

$$q_{t|1,i}(u | a) = (1 - t)q_{0,i}(u) + t \mathbf{1}\{u = a\}, \quad t \in [0, 1], \quad (6)$$

with a univariate conditional rate  $R_{t,i}^*(u, v | a)$  that realizes this path (Qin et al., 2025). FORGE applies this construction independently to the variable coordinates conditional on the shared clean endpoint. Consequently the joint conditional path is a product of coordinate paths, not the mixture  $(1 - t)q_0 + tq_1$ . Its continuous-time generator permits one coordinate

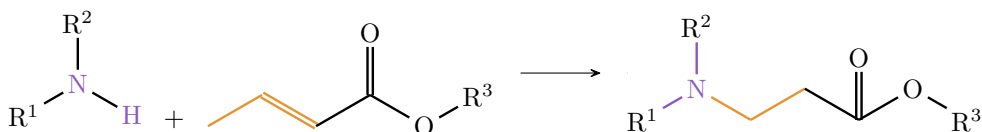
### Ugi three-component reaction

*amine head, aldehyde and isocyanide combine in one step; no carboxylic-acid reactant component*



### Repeated aza-Michael addition

*the head N-H adds across the acrylate double bond; the ester is untouched*



### Repeated reductive amination

*one new carbon-nitrogen bond; the carbonyl oxygen leaves as water*

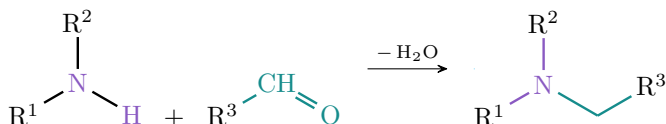


Figure 3: **The three assembly chemistries.** Skeletal notation, with each bond carrying the colour of the component that contributes it: amine head in purple, aldehyde-derived atoms in teal, the isocyanide or acrylate tail in orange. Grey marks an atom the assembly template introduces and no precursor supplies, following the same convention as the core inset of Figure 1. Hydrogens are drawn only on the atoms whose substitution decides whether a product can be decomposed back into precursors: the head N-H, and the  $\alpha$ -carbon C-H that the Ugi reaction takes from the aldehyde.  $R^i$  stand for the source-linked components of the frozen library, which are larger than the fragments drawn here. The Ugi variant has no carboxylic-acid reactant component.

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jump at an infinitesimal time. The denoiser predicts the clean posterior marginal of each coordinate, which is exactly the quantity needed to marginalize the corresponding local rate. Appendix B defines the full path, learned generator and the capped independent-coordinate Euler approximation used in code.

## B FORGE IN FULL

This appendix gives the complete formulation summarized in Section 2 of the body.

### B.1 OVERVIEW

We formulate ionizable-lipid design as conditional generation under a specified reaction program, while leaving precursor identities unspecified. Let  $G \in \mathcal{G}$  denote a complete ionizable-lipid graph and  $P \in \mathcal{P}$  a reaction program specifying the assembly chemistry, precursor roles, reaction-core positions, and transformation order. FORGE models the conditional distribution  $p_\theta(G | P)$  without access to precursor identifiers, stored component graphs, or fragment tokens. FORGE is organized into four stages:

1. **Reaction-program specification.**  $P$  defines the assembly chemistry, precursor roles, reaction-core positions, transformation order, and coarse per-role morphology, but not precursor identities.
2. **Whole-graph generation.** A conditional discrete flow generates every atom, bond, and topology state of  $G$  under  $P$ .
3. **Exact assembly verification.** A reverse transform decomposes  $G$  into role-assigned precursors, and the corresponding forward transform must reconstruct  $G$  exactly.
4. **Recursive synthesis-program construction.** FORGE searches for preparation routes to generated precursors that are not already supported as available starting materials and abstains when a route cannot be completed.

### B.2 SYNTHESIS-GROUNDED DESIGN TASK

Given a reaction program  $P$ , FORGE must generate a complete molecular graph  $G$  and either construct a complete synthesis program for  $G$  or abstain. Let  $\mathfrak{D}$  denote the dossier space and let  $\mathfrak{F}_{\mathfrak{B}}$  be the finite set of typed failure outcomes under a frozen route contract  $\mathfrak{B}_{\text{route}}$ . We write the design process as

$$G \sim p_\theta(\cdot | P), \quad Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (7)$$

$\text{Assess}_{\mathfrak{B}}$  denotes the bounded program constructor, invoked after the verifier selects the sole admitted root trace  $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_P(G)$ , that verifies the final assembly, searches for preparation routes to unavailable precursors, and checks terminal supplier evidence.  $P$  and the assessor act at different stages:  $P$  guides generation, whereas the assessor constructs a dossier from the completed graph without modifying it. FORGE accepts  $(G, T)$  only when  $G$  passes molecular and exact-assembly verification and  $T$  traces every required precursor to an available starting material; otherwise, it abstains.

### B.3 REACTION-PROGRAM SPECIFICATION

**Definition B.1** (Reaction program). *A reaction program is a tuple*

$$P = (c, d, \mathbf{r}, \mathbf{k}, M), \quad (8)$$

*with an ordered layout index set  $\mathcal{I}(P) = \{1, \dots, n(P)\}$ . Here  $c$  identifies the ordered assembly chemistry,  $d$  is the program depth,  $\mathbf{r} = (r_i)_{i \in \mathcal{I}(P)}$  assigns precursor roles,  $\mathbf{k} = (k_i)_{i \in \mathcal{I}(P)}$  marks reaction-core positions, and  $M$  specifies coarse per-role morphology. For each precursor role  $r$ ,  $M$  records four integers: the number of exterior atoms  $n_r$ , junction budget  $j_r$ , cycle rank  $\gamma_r$ , and attachment count  $a_r$ .*

$M$  is defined relative to an exact atom-mapped trace  $\tau$  and precursor tuple  $\mathbf{C}$ . An atom is exterior for role  $r$  when  $\tau$  assigns it to the corresponding precursor and does not mark it as reaction core. Template-introduced atoms belong to the core and to no precursor role. The role-induced subgraph contains the atoms assigned to  $r$  and all product bonds with

both endpoints in that set. Its junction count is  $\sum_v \max\{\deg_r(v) - 2, 0\}$ , its cycle rank is  $|E_r| - |V_r| + \kappa_r$  for  $\kappa_r$  connected components, and its core-attachment count is the number of product bonds between an exterior atom of  $r$  and a core atom. We write  $(G, \mathbf{C}, \tau) \models M(P)$  when these values match the sampled counts, with junction count at most its budget. Thus morphology is decomposition-relative rather than an unlabeled property of  $G$ .

FORGE first samples an admissible complete program  $P$  from the training-fold program prior and then generates  $G$  conditioned on it:

$$p_\theta(G, P \mid c) = p_{\text{prog}}(P \mid c) p_\theta(G \mid P). \quad (9)$$

The prior sets each region’s atom count, branching budget, cycle rank, and number of core attachments, but not its molecular structure. A chemistry-specific adapter defines the precursor roles, reaction core, reverse decomposition, and forward transform associated with  $c$ . For each graph position  $i$ , FORGE combines learned embeddings of the program identity, program depth, precursor role, and reaction-core position:

$$h_i^P = \text{LN} \left( e_c(c) + e_d(d) + e_r(r_i) + e_k(k_i) + \sum_{u=1}^4 e_{M,u}(M_{r_i,u}) \right). \quad (10)$$

The morphology fields also determine active positions and fixed/variable masks in the sampling layout. These coordinates tell the generator which precursor role each position plays, whether it lies in the reaction core, and the role-local count constraints, but not which molecular structure should occupy that role.

#### B.4 PROGRAM-DEFINED MOLECULAR SPACE

Reaction programs constrain how a lipid is assembled without specifying the molecular structures that occupy its precursor roles. Let  $\mathcal{G}_{\text{bounded}} \subseteq \mathcal{G}$  denote the molecular graphs within the declared atom and bond vocabularies, the 194-heavy-atom limit, and the three-closure limit. For chemistry  $c$ , let  $\mathcal{D}_c(G)$  be the candidate exact-decomposition search space and define

$$\mathcal{D}_P(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_c(G) : (G, \mathbf{C}, \tau) \models P\}. \quad (11)$$

The valid, exact-L1 molecular space associated with  $P$  is

$$\mathcal{G}_{\text{L1}}(P) = \left\{ G \in \mathcal{G}_{\text{bounded}} : \text{Valid}(G) = 1, \widehat{\text{L1}}(G, P) = 1 \right\}. \quad (12)$$

A generated lipid can pass the exact L1 check even when one or more recovered precursors never appeared in the training component set, provided the specified reaction reconstructs the lipid from them. For a reaction program  $P$ , let  $\mathfrak{C}_P$  denote its admissible precursor-program inputs, including precursor identities, role assignments, and program depth. The registered forward operator maps each input to a finite set of canonical products:

$$\mathcal{F}_P : \mathfrak{C}_P \longrightarrow 2^{\mathfrak{G}^{N, K_{\text{cl}}}}. \quad (13)$$

Given role-specific catalogues  $\mathcal{C}$ , let  $\mathfrak{C}_P(\mathcal{C}) \subseteq \mathfrak{C}_P$  contain the admissible program inputs whose components all belong to their corresponding catalogues. The catalogue-reachable product set is

$$\mathcal{G}_{\text{cat}}(P, \mathcal{C}) = \bigcup_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} \mathcal{F}_P(\mathbf{C}). \quad (14)$$

For any graph  $G$ , define its complete program-relative preimage as

$$\mathcal{E}_P^*(G) = \{\mathbf{C} \in \mathfrak{C}_P : G \in \mathcal{F}_P(\mathbf{C})\}. \quad (15)$$

**Definition B.2** (Strong catalogue escape).

$$\text{Escape}^*(G; P, \mathcal{C}) = \mathbf{1}[\mathcal{E}_P^*(G) \neq \emptyset \wedge \mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C}) = \emptyset]. \quad (16)$$

Proposition E.3 bounds the catalogue-reachable product set. For a verified exact-L1 graph, direct membership in the canonical product-hash set gives the operational strong-escape test

$$\text{StrongEscape}(G; P, \mathcal{C}) = \mathbf{1}[\widehat{\text{L1}}(G, P) = 1 \wedge G \notin \mathcal{G}_{\text{cat}}(P, \mathcal{C})]. \quad (17)$$

The verifier-returned set  $\widehat{\mathcal{E}}_P(G)$  defined below is a subset of  $\mathcal{E}_P^*(G)$  when every returned trace passes exact forward replay. A novel component in a returned trace establishes strong catalogue escape only when reverse enumeration is exhaustive or direct catalogue reachability is checked independently. The present experiments did not precompute a complete common-reference product-hash set, so they report verifier-recovered component novelty as a secondary diagnostic and do not report StrongEscape for FORGE. FORGE instead parameterizes  $p_\theta(G \mid P)$  over whole graphs in  $\mathcal{G}_{\text{bounded}}$ , without including any component catalogue  $\mathcal{C}_r$  in the generative state. Reaction programs constrain assembly semantics, while the flow determines precursor identity at atom and bond resolution.

### B.5 WHOLE-GRAPH GENERATION

FORGE realizes this whole-graph distribution with a discrete flow over atom, bond, and topology states. To preserve the full 194-atom support without an  $O(N_{\text{max}}^2)$  edge state, FORGE encodes each graph as a spanning tree plus at most three residual closure edges. Let  $\alpha = (\mathbf{C}, \tau, \mathbf{r}, \mathbf{k})$  contain the selected exact precursor tuple, atom-mapped reaction trace, precursor-role assignment and reaction-core annotation. The deterministic program-aware encoder

$$\text{Enc}_P : \{(G, \alpha) : G \in \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \text{ and } \alpha \text{ is admitted for } P\} \longrightarrow \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \quad (18)$$

constructs the categorical graph state

$$X_G := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}) \quad (19)$$

where  $\mathbf{a}$  contains atom states,  $\boldsymbol{\pi}$  parent pointers,  $\mathbf{b}$  spanning-tree bond states,  $\mathbf{e}^L$  and  $\mathbf{e}^R$  closure-endpoint pointers, and  $\mathbf{z}$  closure-bond states.

The encoder first partitions the atoms into connected precursor-origin components under  $\alpha$ . It chooses as root the reaction-core atom with the smallest tied-broken canonical RDKit rank, and orders the origin-component graph by breadth-first traversal from that root component, breaking ties by role index, component canonical ranks and original atom index. Within each component it uses a deterministic depth-first preorder from the selected attachment atom, with neighbors ordered by the same canonical ranks. The traversed bonds define the spanning tree. Every residual bond is represented once as an ordered endpoint pair with its bond label, and residual triples are sorted lexicographically. Role, core, fixed-coordinate and morphology arrays are permuted by precisely the same atom order. Training records with more than one qualified exact source trace are not resolved arbitrarily: they abstain from semantic training; duplicate descriptions of the same admitted trace collapse to its canonical record. Thus  $\text{Dec}(\text{Enc}_P(G, \alpha)) \cong G$  for every admitted record.

The raw padded categorical product space  $\overline{\mathfrak{X}}_{N_{\text{max}}, K_{\text{cl}}}$  also contains malformed states. Its well-formed subset  $\mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}}$  enforces an active node prefix, earlier-parent pointers, distinct valid closure endpoints, no duplicate edges, and null values in unused closure slots. The partial decoder is therefore typed as

$$\text{Dec} : \overline{\mathfrak{X}}_{N_{\text{max}}, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (20)$$

For this study,  $N_{\text{max}} = 194$  and  $K_{\text{cl}} = 3$ . This is a statement about architectural support. It does not imply reaction consistency, nonzero probability under a trained model, or synthesis-program completeness.

If  $q_\theta(x \mid P)$  is the terminal raw-state law, the generated graph distribution is its decoder pushforward,

$$p_\theta([G]_{\cong} \mid P) = \sum_{x: \text{Dec}(x) \cong G} q_\theta(x \mid P), \quad p_\theta(\perp_{\text{dec}} \mid P) = \sum_{x: \text{Dec}(x) = \perp_{\text{dec}}} q_\theta(x \mid P). \quad (21)$$

The chemistry adapter marks any atom, pointer, or bond states fixed by  $P$ . FORGE holds these states constant while corrupting the remaining states along the linear probability path and predicting their clean values from  $(X_t, t, P)$ . Let  $\Omega_P$  be the face of the raw state space satisfying those fixed coordinates, and define the admissible program set

$$\mathcal{P}_{\text{adm}} = \{P : \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}. \quad (22)$$

The morphology prior is normalized only on  $\mathcal{P}_{\text{adm}}$ . Each chemistry adapter is required to satisfy the program-compatible encoding property: every admitted annotated record obeys  $\text{Enc}_P(G, \alpha) \in \Omega_P$ . Corpus construction checks this invariant together with exact constitutional decode round trip and fails closed before a record enters any split. This is an exhaustive implementation invariant over the cached training, calibration and held-out records, not a theorem for unregistered chemistries.

Let  $\mathcal{J} = \{\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}\}$  denote the six state families optimized during training. Let  $\mu_{\text{bal}}$  denote the frozen training distribution over graph-program pairs, which assigns equal total mass to each reaction program and preserves the source-balanced weights within each program. For each  $q \in \mathcal{J}$ , let  $m_j^{(q)}(P) \in \{0, 1\}$  indicate whether coordinate  $j$  remains variable under  $P$ . FORGE minimizes a program-balanced denoising objective over the variable coordinates:

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G, P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0, 1), X_t \sim p_t(\cdot | X_G, P)} \left[ \sum_{q \in \mathcal{J}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \left[ -\log p_\theta \left( X_{G,j}^{(q)} | X_t, t, P \right) \right]}{\max \left\{ 1, \sum_j m_j^{(q)}(P) \right\}} \right]. \quad (23)$$

Here,  $\lambda_q$  weights state family  $q$ , and a family with no variable coordinates contributes zero loss. The corruption and sampling operators assign zero rates to program-fixed coordinates and project each finite-step state back onto them as an implementation safeguard. This is the precise syntactic sense in which the program constrains a state trajectory: fixed coordinates cannot be resampled. Chemical consistency with the requested assembly is established separately by exact forward replay. Under an unrestricted coordinatewise model class, every population minimizer of the primary masked denoising term recovers the corresponding clean-coordinate posterior marginals. This result does not characterize the optimizer of the auxiliary-loss or PCGrad procedure (Propositions E.5 and E.6).

The denoiser is a graph Transformer over the complete sparse state. Each layer applies graph self-attention with relation biases for parent, child and closure edges, then cross-attends to memory tokens representing program identity, program depth, precursor role and reaction-core position. Cross-attention is repeated in every layer so that reaction semantics can influence both local state updates and long-range coordination among precursor-derived regions. Three softly program-routed residual adapters provide family-specific capacity while retaining a shared whole-graph backbone.

For each reaction family  $f$ , we compute a task loss

$$\mathcal{L}_f = \mathcal{L}_{\text{FORGE},f} + \lambda_{\text{role}} \mathcal{L}_{\text{role}} + \lambda_{\text{core}} \mathcal{L}_{\text{core}}. \quad (24)$$

Present semantic states receive equal mass within each auxiliary term. Training gives every reaction family equal total loss mass. The gradients of the family task losses, not the individual loss heads, are combined by deterministic PCGrad. Families are ordered by their integer program state. Writing  $g_f = \nabla_\theta \mathcal{L}_f$ , the algorithm initializes  $\tilde{g}_f \leftarrow g_f$  and, for every other family  $h$  in that fixed order, applies

$$\tilde{g}_f \leftarrow \tilde{g}_f - \mathbf{1}\{\langle \tilde{g}_f, g_h \rangle < 0\} \frac{\langle \tilde{g}_f, g_h \rangle}{\max(\|g_h\|_2^2, 10^{-12})} g_h \quad (25)$$

on parameter coordinates available to both tasks. Zero gradients remain zero; gradients are not norm-normalized; and the optimizer receives the arithmetic mean of the projected family gradients. This is an algorithmic definition of the update and is not the gradient of the displayed scalar sum in general. Table 7 prespecifies separate interventions for layerwise cross-attention, each auxiliary loss, routed adapters and gradient conflict control.

At inference, FORGE samples  $X_0$  from the factorized program-compatible source, assigns zero update rates to fixed coordinates, and applies  $L_{\text{samp}}$  capped independent-coordinate Euler steps. At each step the implementation samples one clean category per coordinate from the denoiser, forms the minimum  $R^*$  rate for that sampled endpoint, multiplies it by  $\Delta t = 1/L_{\text{samp}}$ , and rescales the total off-diagonal mass to at most 0.999 before drawing

the next coordinate value. This is a finite-step Monte Carlo approximation to the local CTMC, not exact CTMC simulation. Appendix E states the ideal marginalized rate and the numerical kernel. FORGE decodes the terminal state  $X_1$  into a molecular graph  $G$  and retains it only if it is connected, valence-valid, and passes molecular sanitization.

## B.6 EXACT ASSEMBLY VERIFICATION

To ensure that each retained graph satisfies the specified reaction program, FORGE decomposes it into role-assigned precursors and applies the corresponding forward transform.

**Definition B.3** (Mathematical and verifier-relative L1). *For chemistry  $c$ , the reverse search  $\mathcal{D}_c$  maps  $G$  to candidate role-assigned precursor tuples and atom-mapped traces. The complete mathematical preimage  $\mathcal{E}_P^*(G)$  contains every admissible input whose registered forward operator can produce  $G$ . Define*

$$\text{L1}^*(G, P) = \mathbf{1}[\mathcal{E}_P^*(G) \neq \emptyset]. \quad (26)$$

*The bounded implementation searches  $\mathcal{D}_P(G) \subseteq \mathcal{D}_c(G)$  and returns*

$$\widehat{\mathcal{E}}_P(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}) \text{ such that } H \cong G\}. \quad (27)$$

*Its verified predicate is*

$$\widehat{\text{L1}}(G, P) = \mathbf{1}[\widehat{\mathcal{E}}_P(G) \neq \emptyset]. \quad (28)$$

Every returned trace passes exact replay, hence  $\widehat{\text{L1}}(G, P) \leq \text{L1}^*(G, P)$ . Reverse-search completeness has not been proved, so the reported metric is called *verified exact-L1 yield*. Post-hoc filtering changes which completed samples are retained but cannot change the proposal distribution’s exact-L1 acceptance probability. Under independent sampling, this probability determines the complete distribution of accepted counts under a fixed attempt budget and the expected cost of obtaining a fixed number of accepted products (Proposition E.9). FORGE reports decomposition coverage separately from forward-replay precision because identifying a candidate disconnection does not guarantee exact reconstruction of  $G$ . Candidate decomposition coverage is the fraction of valid graphs for which  $\mathcal{D}_c(G) \neq \emptyset$ . Candidate replay precision is

$$\text{ReplayPrecision} = \frac{\sum_G \sum_{(\mathbf{C}, \tau) \in \mathcal{D}_P(G)} \mathbf{1}\{\exists H \in \mathcal{F}_P(\mathbf{C}) : H \cong G\}}{\sum_G |\mathcal{D}_P(G)|}, \quad (29)$$

with N/E when the denominator is zero. Replay of tuples already in  $\widehat{\mathcal{E}}_P(G)$  is a 100% verifier invariant rather than an empirical precision result. We separately define

$$\text{CandidateAmbig}(G, c) = \mathbf{1}\{|\mathcal{D}_c(G)| > 1\}, \quad \text{ExactAmbig}(G, P) = \mathbf{1}\{|\widehat{\mathcal{E}}_P(G)| > 1\}. \quad (30)$$

A distinct exact-L1 product is a deduplicated generated constitution; an unambiguous exact decomposition instead has  $|\widehat{\mathcal{E}}_P(G)| = 1$ . We do not use “unique L1” for both.

## B.7 RECURSIVE SYNTHESIS-PROGRAM CONSTRUCTION

The final assembly requires access to every recovered precursor. Exact L1 verification identifies these precursors, but it does not establish whether they are available or how to prepare them. For laboratory planning, an exact final assembly is useful only when every required precursor is either available or connected to available starting materials through a supported preparation route. Let  $T = (V_M, V_R, E)$  denote a bipartite synthesis tree rooted at  $G$ , where each product molecule points to its preparation reaction and each reaction points to its required reactants. FORGE constructs these routes recursively and rejects the lipid when any branch remains unresolved. A molecule node terminates when a current, dated supplier record lists an exact constitutional match as available; otherwise, FORGE must expand that node through a supported, forward-verified reaction. FORGE uses a bounded level-two (L2) planner to propose each expansion. For each proposal, FORGE records the reaction, reactants, product, and supporting evidence, and accepts the step only when

the forward transform reconstructs the target precursor exactly. Each accepted reaction adds its reactants as new molecule nodes, and FORGE applies the same termination-or-expansion rule recursively. The current assessor admits only an unambiguous verified exact-L1 root. Every dossier therefore records the sole selected root trace  $(\mathbf{C}_T, \tau_T) \in \widehat{\mathcal{E}}_P(G)$ ; an ambiguous exact set produces a typed abstention. For each upstream reaction node  $r$ , let  $\text{Admitted}_{\mathfrak{B}}(r) = 1$  when its reaction and substrate scope satisfy the frozen evidence policy, and let  $\text{Verify}_{\mathfrak{B}}(r) = 1$  when  $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$ . For each leaf molecule  $\ell$ , let  $\text{Available}_{\mathfrak{B}}(\ell) = 1$  when it satisfies the terminal supplier-evidence rule. Let  $V_R^{\text{up}}(T)$  denote the upstream preparation-reaction nodes and  $L(T)$  the leaf molecules.  $\text{WellFormed}(G, T, P)$  requires that  $T$  is a connected, acyclic bipartite tree rooted at  $G$ ; each internal molecule has exactly one recorded producing reaction; each reaction’s incoming molecule children equal its recorded reactants; and  $T$  has no disconnected, cyclic or extraneous node. Its distinguished final-assembly record must have product  $G$  and precursor children exactly  $\mathbf{C}_T$  under trace  $\tau_T$ .

$$\begin{aligned} \text{Complete}_{\mathfrak{B}}(G, T, P) = & \text{WellFormed}(G, T, P) \mathbf{1}[(\mathbf{C}_T, \tau_T) \in \widehat{\mathcal{E}}_P(G)] \\ & \times \prod_{r \in V_R^{\text{up}}(T)} \text{Admitted}_{\mathfrak{B}}(r) \text{Verify}_{\mathfrak{B}}(r) \prod_{\ell \in L(T)} \text{Available}_{\mathfrak{B}}(\ell). \end{aligned} \quad (31)$$

Empty products equal one. Let  $\mathfrak{B}_{\text{route}}$  fix the route-depth and query budgets, registered transforms, evidence-admission policy, and dated supplier snapshot used by  $\text{Assess}_{\mathfrak{B}}$ . Under finite search bounds and terminating evidence queries, the assessor terminates and returns a dossier only when  $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$  (Lemma E.10 and Invariant E.11). An abstention means that FORGE did not complete a dossier under  $\mathfrak{B}_{\text{route}}$ ; it does not imply that no synthesis route exists outside the frozen search space or evidence snapshot. The guarantee is computational and does not establish experimental synthesis success.

Algorithm 1 summarizes the complete inference procedure. Every iteration consumes one generation attempt, and FORGE retains the outcome of every failed attempt in a typed ledger.

For every accepted pair  $(G, T)$ , FORGE records a computational synthesis dossier containing all molecules and reactions in  $T$ , the evidence supporting each reaction, and the dated supplier record for every terminal material. We classify each generated lipid into one of four support tiers according to which elements of its assembly lie outside the fixed reference space: **E0 (enumerated)**: the generated lipid matches a product that was already constructed computationally from known components using a supported reaction. **E1 (known components)**: the generated lipid was not included in the bounded E0 enumeration, but a supported reaction assembles it entirely from known components. **E2 (generated precursor)**: a supported reaction assembles the lipid, but at least one precursor is newly generated and requires a complete L2 route to available starting materials. **E3 (new assembly family)**: the lipid requires a final assembly reaction outside the qualified reaction registry.

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**Algorithm 1** FORGE generation and fail-closed synthesis-program construction.

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**Require:** trained coordinate predictors, reaction program  $P$ , Euler-step count  $L_{\text{samp}}$ , attempt budget  $B$ , recorded seed  $s$ , and frozen route contract  $\mathfrak{B}_{\text{route}}$

**Ensure:** attempt-indexed dossier sequence  $\mathcal{D}_{\text{out}}$ , exact-L1 ledger  $\mathcal{L}_{\text{out}}$ , and typed failure ledger  $\mathcal{H}$

```

1: initialize the random-number streams from  $s$ ;  $\mathcal{D}_{\text{out}} \leftarrow []$ ;  $\mathcal{L}_{\text{out}} \leftarrow []$ ;  $\mathcal{H} \leftarrow []$ 
2: for  $b = 1, \dots, B$  do
3:   sample  $X_0 \sim q_0^P$  and set  $X \leftarrow \Pi_P(X_0)$ 
4:   for  $k = 0, \dots, L_{\text{samp}} - 1$  do
5:      $t_k \leftarrow k/L_{\text{samp}}$ ;  $\Delta t_k \leftarrow 1/L_{\text{samp}}$ 
6:      $X \leftarrow \Pi_P(\text{CAPPEDCOORDINATEEULER}(X, t_k, \Delta t_k, P))$ 
7:   end for
8:    $G \leftarrow \text{DECODE}(X)$ 
9:   if  $G = \perp_{\text{dec}}$ , is valence-invalid, or fails molecular sanitization then
10:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{INVALIDGRAPH})]$ 
11:    continue
12:   end if
13:    $\hat{\mathcal{E}}_P(G) \leftarrow \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}), H \cong G\}$ 
14:   if  $\hat{\mathcal{E}}_P(G) = \emptyset$  then
15:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{L1FAILURE})]$ 
16:    continue
17:   end if
18:    $\mathcal{L}_{\text{out}} \leftarrow \mathcal{L}_{\text{out}} \parallel [(b, G, \hat{\mathcal{E}}_P(G))]$ 
19:   if  $|\hat{\mathcal{E}}_P(G)| \neq 1$  then
20:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{AMBIGUOUSL1})]$ 
21:    continue
22:   end if
23:   select the sole  $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_P(G)$ 
24:    $Y \leftarrow \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$ 
25:   if  $Y$  is a tree  $T$  and  $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$  then
26:     $\mathcal{D}_{\text{out}} \leftarrow \mathcal{D}_{\text{out}} \parallel [(b, \text{DOSSIER}(G, P, \mathbf{C}_T, \tau_T, T))]$ 
27:   else
28:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, Y)]$   $\triangleright$  preserve the typed outcome
29:   end if
30: end for
31: return  $\mathcal{D}_{\text{out}}, \mathcal{L}_{\text{out}}, \mathcal{H}$ 

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## C EXTENDED EXPERIMENTAL PROTOCOL AND RESULTS

This appendix gives the full reporting contract, including every matched arm, ablation and diagnostic summarized in Section 3 of the body.

### C.1 EXPERIMENTAL SETTINGS

Our evaluation spans the three assembly programs defined above, with Ugi as the deeply supervised case, repeated aza-Michael addition as a second family, and repeated reductive amination as a lighter stress test. The evaluation tests four capabilities: learning multiple assembly programs within one model, using reaction semantics during generation rather than after it, recovering verified decompositions with components absent from the training catalogue, and tracing generated precursors to available starting materials. We compare four matched training arms: Ugi-only conditioning, shared conditioning across all three programs, shared training without program information, and shared training with cyclically incorrect program labels. All arms use the same architecture and optimization budget, and we evaluate the prespecified final checkpoint from each of three independent seeds. Each matched comparison uses the same number of generation attempts per program. Our primary metric is verified exact-L1 yield per generation attempt: the number of generated graphs that pass molecular validity, program-aware decomposition, and exact forward replay divided by the total number of sampling attempts. Secondary metrics separate graph quality (validity and connectedness), reaction recovery (verified coverage, the replay invariant, abstention, and exact ambiguity), and exploration (distinct-product fraction, whole-product and component novelty, molecular diversity, and effective component count). We report exact counts, means across seeds, sample standard deviations, individual seed values, and the minimum and maximum of the three paired seed contrasts. These are descriptive across-seed summaries, not confidence intervals. Generated molecules are not treated as independent training replicates. Conditional binomial variation across a fixed model’s 3,072 attempts is distinct from variation induced by retraining. Training uses only the component-disjoint training folds. Calibration data are diagnostic, and held-out products neither enter training nor select checkpoints. The three shared arms assign equal total sampling mass to each program while preserving the frozen within-program source weights. The frozen implementation evaluated numerical promotion gates, but three seeds do not support a population-level non-inferiority claim. We therefore report only whether the observed runs met each numeric rule. Zero program-fixed-coordinate violations are an implementation check required by Proposition E.5, not learned behavior.

Two additional comparisons isolate the role of reaction semantics. First, we compare conditioned generation with null generation followed by exact post-hoc reaction filtering under the same attempt budget; the cyclic-label arm tests whether chemical meaning matters beyond the presence of a categorical input. Second, we compare FORGE with a train-only finite-component assembler that receives the exact training components, supported forward chemistry, source-weighted role marginals, and the same number of attempts. We report reaction validity and exploration separately because the catalogue can achieve high exact-L1 yield while having zero input-component novelty by construction. These comparisons make no L2 planner, biological-oracle, or candidate-selection calls.

For the Ugi program, we additionally define a common external-baseline protocol with one reaction-space comparator, RGFN, and two generic whole-molecule comparators, unconditional DeFoG and GenMol/SAFE (Koziarski et al., 2024; Qin et al., 2025; Lee et al., 2025). The Synthesis-DAG method of Ou et al. (2024), SynFlowNet and SynCoGen remain methodological context rather than empirical rows (Cretu et al., 2025; Reakes et al., 2025). The released Synthesis-DAG repository does not contain the paper-specific ionizable-lipid implementation, the released SynFlowNet action space cannot represent the qualified three-reactant Ugi transform, and the released SynCoGen repository has no usable code license. We do not replace missing functionality with an imitation, alter Ugi into sequential pseudo-reactions, or execute unlicensed code. An external row enters the numerical comparison only after the method has been executed on the same training split, generation-attempt budget, seed set, and common molecular and exact-L1 evaluation pipeline used for FORGE. Published values obtained on the method’s original dataset are not mixed with results from

this common evaluation. Methods that do not support the additional assembly programs are compared on Ugi only rather than assigned artificial aza-Michael or reductive-amination results. The matched internal suite separately includes unconditioned generation plus post-hoc filtering, an oracle finite-catalogue assembler, a learned inventory selector, two learned component-factorized generators, and targeted interventions on the Transformer. Table 7 reports the completed three-seed Transformer mechanism and component-factorization study. Each intervention uses the same folds, generation budget and evaluation code as the full Transformer.

## C.2 METRIC, IDENTITY AND UNCERTAINTY DEFINITIONS

Let  $Y_{s,a,b}^{\text{L1}} \in \{0, 1\}$  indicate verified exact L1 for training seed  $s$ , arm  $a$  and attempt  $b$ , and let  $B = 3,072$ . We report the exact count  $C_{s,a}^{\text{L1}} = \sum_{b=1}^B Y_{s,a,b}^{\text{L1}}$  and

$$\hat{p}_{s,a}^{\text{L1}} = C_{s,a}^{\text{L1}}/B. \quad (32)$$

Validity and connected yield replace  $Y^{\text{L1}}$  with their attempt-level indicators. Verified decomposition coverage is the number of valid graphs with  $\hat{\mathcal{E}}_P(G) \neq \emptyset$  divided by valid graphs; the implementation does not report candidate-search coverage before replay. Exact ambiguity is the fraction of valid graphs with  $|\hat{\mathcal{E}}_P(G)| > 1$ . Replay of members of  $\hat{\mathcal{E}}_P(G)$  is an invariant, while candidate replay precision is defined in Appendix B and is N/E if no candidate tuple is returned. Explicitly, with attempt outcomes  $G_{s,a,b}$  and the convention that a failed decode makes every graph predicate zero,

$$\hat{p}_{s,a}^{\text{valid}} = B^{-1} \sum_b \mathbf{1}\{\text{Valid}(G_{s,a,b}) = 1\}, \quad (33)$$

$$\hat{p}_{s,a}^{\text{conn}} = B^{-1} \sum_b \mathbf{1}\{\text{Valid}(G_{s,a,b}) = 1, \text{Connected}(G_{s,a,b}) = 1\}, \quad (34)$$

$$\text{CandidateCoverage} = \frac{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1, \mathcal{D}_c(G_b) \neq \emptyset\}}{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1\}}, \quad (35)$$

$$\text{ExactAmbiguity} = \frac{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1, |\hat{\mathcal{E}}_P(G_b)| > 1\}}{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1\}}. \quad (36)$$

Each ratio is N/E when its denominator is zero.

Distinct-product yield is the number of distinct canonical product constitutions satisfying verified exact L1 divided by  $B$ . The reported verifier-recovered component-novel yield additionally requires  $|\hat{\mathcal{E}}_P(G)| = 1$  and at least one returned role-assigned precursor constitution absent from the fixed training catalogue. It is not StrongEscape. Held-component recovery counts verified exact-L1 attempts whose returned tuple contains a designated held constitution and divides by  $B$ . Writing  $\text{can}(G)$  for constitutional canonicalization,  $\mathcal{C}_{\text{train}}$  for the role-indexed training catalogue and  $\mathcal{H}_{\text{held}}$  for the designated held constitutions, these yields are

$$\text{DistinctL1Yield} = B^{-1} \left| \left\{ \text{can}(G_b) : \widehat{\text{L1}}(G_b, P) = 1 \right\} \right|, \quad (37)$$

$$\text{ComponentNovelYield} = B^{-1} \sum_b \mathbf{1} \left[ \widehat{\text{L1}}(G_b, P) = 1, |\hat{\mathcal{E}}_P(G_b)| = 1, \right. \\ \left. \exists (r, C) \in \mathbf{C}_b : C \notin \mathcal{C}_{\text{train},r} \right], \quad (38)$$

$$\text{HeldComponentYield} = B^{-1} \sum_b \mathbf{1} \left[ \widehat{\text{L1}}(G_b, P) = 1, \mathbf{C}_b \cap \mathcal{H}_{\text{held}} \neq \emptyset \right]. \quad (39)$$

For component frequencies  $w_i$ , the effective count is

$$N_{\text{eff}} = \exp \left( - \sum_i w_i \log w_i \right). \quad (40)$$

Component duplicates are retained when estimating  $w_i$ . Molecular diversity is the mean pairwise ECFP4 Tanimoto distance among distinct valid canonical product constitutions, so molecular duplicates are removed for that metric. For distinct products  $g_1, \dots, g_m$ ,

$$\text{Diversity} = \frac{2}{m(m-1)} \sum_{1 \leq i < j \leq m} [1 - \text{Tanimoto}(\text{ECFP4}(g_i), \text{ECFP4}(g_j))], \quad (41)$$

with N/E for  $m < 2$ . Every other undefined conditional denominator is likewise reported N/E.

Identity is constitutional and stereo-free. Graph and product identity use connected canonical RDKit SMILES with atom maps and stereochemical tags removed; formal charges and bond orders are retained; salts or disconnected products are invalid; and no tautomer normalization is applied. Precursor identity uses the same canonicalization. Precursor-tuple equality is role- and program-step-aware, with repeated inputs compared in their registered step order. Atom mappings and the trace  $\tau$  are evidence for semantic alignment, not part of constitutional identity.

For a paired contrast,  $D_s = \hat{p}_{s,\text{FORGE}} - \hat{p}_{s,\text{control}}$ . We report  $(D_1, D_2, D_3)$ ,  $\bar{D}$ , and  $[\min_s D_s, \max_s D_s]$ . These summarize the three observed training seeds and are not 95% confidence intervals. Conditional on one fixed trained model,  $C_{s,a}^{\text{L1}}$  has the finite-attempt binomial variation in Proposition E.9; that uncertainty does not substitute for independent retraining. Unless explicitly named primary, architecture, coordinate, guidance, potency and catalogue contrasts are exploratory and no multiplicity-adjusted inferential claim is made.

The frozen Ugi-retention contract had numerical margins on the challenger relative to Ugi-only: maximum absolute drops of 2 percentage points for validity, 1 point for connectedness, 3 points for verified decomposition coverage, 1 point for replay precision with an absolute floor of 95%, 5 points for component novelty, 5 points for held-component decomposition coverage and 0.05 for diversity;  $N_{\text{eff}}$  had a minimum ratio of 0.8; fixed-coordinate failures and support overflows had maximum zero. The original artifact applied one-sided bootstrap bounds, but with three seeds we use only the observed numerical checks and do not interpret them as population non-inferiority tests. Formally, a future non-inferiority analysis would prespecify  $H_0 : \Delta \leq -\delta$  versus  $H_1 : \Delta > -\delta$ , the one-sided level, independent-seed count and multiplicity rule before training.

### C.3 SHARED REACTION-PROGRAM LEARNING ACROSS THREE ASSEMBLY CHEMISTRIES

Across 3,072 held-out generation attempts per program in each of three independent training seeds, the final conditioned model achieved verified exact-L1 yields of  $78.7 \pm 10.1\%$  for Ugi (72.3%, 90.3% and 73.5% by seed),  $77.5 \pm 3.7\%$  for repeated aza-Michael addition (76.5%, 81.5% and 74.3%), and  $31.3 \pm 14.3\%$  for repeated reductive amination (38.6%, 14.8% and 40.4%) (mean  $\pm$  sample standard deviation; Table 1 and Table 4). Every verifier-returned trace passed forward replay, as required by the verifier invariant. However, mean exact-decomposition ambiguity was 54.9% for repeated aza-Michael addition and 74.1% for repeated reductive amination, and the latter program showed substantial between-seed variation. Table 4 reports molecular validity, connectedness, decomposition coverage, forward-replay precision, abstention, ambiguity, novelty and diversity separately for each program.

Relative to the Ugi-only model, shared training changed verified Ugi exact-L1 yield by 53.26 percentage points. The paired seed differences were 41.24, 56.71, 61.82 points, spanning 41.24 to 61.82, and the observed runs met the frozen numerical retention rules. The final conditioned model recorded zero fixed-coordinate violations and zero support overflows across 27,648 held-out attempts (three seeds, three programs and 3,072 attempts per program). Together, these results support the conclusion that one conditional flow can learn the three specified reaction-program coordinates without erasing the deeply supported Ugi behavior.

Table 4: **Matched production comparison across reaction programs.** Point estimates report the mean across three independent seeds; the exact-L1 column additionally reports the sample standard deviation. Paired contrasts are summarized by their three seed values, mean and observed range; individual seed values appear in the supplement. Validity and connectedness use all generation attempts as the denominator. Decomposition coverage, replay precision, abstention and ambiguity retain their prespecified conditional denominators. Novelty is conditioned on valid products or unique returned decompositions, as appropriate. Fraction pairs and exact-L1 yield are percentages; diversity is mean pair-wise ECFP4 distance and  $N_{\text{eff}}$  is the effective generated-component count. Results are never pooled across reaction programs.

|                             | Ugi             |                 |                |                | Aza-Michael    |               |               | Reductive amination |                |                |
|-----------------------------|-----------------|-----------------|----------------|----------------|----------------|---------------|---------------|---------------------|----------------|----------------|
|                             | Ugi-only        | FORGE           | Shared null    | Cyclic         | FORGE          | Shared null   | Cyclic        | FORGE               | Shared null    | Cyclic         |
| Valid./Conn.                | 28.0/28.0       | 86.6/86.6       | 19.5/19.5      | 43.4/43.4      | 77.5/77.5      | 16.5/16.5     | 25.7/25.7     | 36.5/36.5           | 15.4/15.4      | 21.4/21.4      |
| Decomp./Replay              | 89.9/100.0      | 90.7/100.0      | 58.8/100.0     | 92.7/100.0     | 100.0/100.0    | 24.3/100.0    | 15.2/100.0    | 84.3/100.0          | 85.4/100.0     | 84.9/100.0     |
| Abst./Ambig.                | 10.1/0.0        | 9.3/0.0         | 41.2/0.0       | 7.3/0.0        | 0.0/54.9       | 75.7/0.1      | 84.8/0.0      | 15.7/74.1           | 14.6/70.7      | 15.1/74.6      |
| Exact-L1/attempt            | 25.4 $\pm$ 12.0 | 78.7 $\pm$ 10.1 | 11.4 $\pm$ 1.1 | 40.2 $\pm$ 0.7 | 77.5 $\pm$ 3.7 | 4.0 $\pm$ 0.8 | 3.8 $\pm$ 0.6 | 31.3 $\pm$ 14.3     | 13.2 $\pm$ 1.3 | 18.1 $\pm$ 0.8 |
| Whole/Comp. novelty         | 99.8/92.4       | 99.8/96.2       | 100.0/99.9     | 99.9/90.1      | 99.7/99.2      | 99.9/99.7     | 99.8/98.3     | 98.1/100.0          | 99.5/100.0     | 96.2/100.0     |
| Diversity/ $N_{\text{eff}}$ | 0.597/134.8     | 0.621/320.8     | 0.754/219.9    | 0.562/138.0    | 0.521/156.9    | 0.635/111.3   | 0.388/51.5    | 0.624/102.9         | 0.615/77.2     | 0.570/34.8     |

Table 5: **Exact verified-L1 counts for the final conditioned model.** Each row reports the numerator, fixed attempt denominator and resulting percentage for one independently trained seed. These attempt-level counts quantify conditional Monte Carlo variation; the three seeds, rather than the individual attempts, are the units for retraining variation.

| Program             | Seed     | Verified L1 count | Attempts | Yield (%) |
|---------------------|----------|-------------------|----------|-----------|
| Ugi                 | 20260825 | 2,220             | 3,072    | 72.27     |
| Ugi                 | 20260826 | 2,774             | 3,072    | 90.30     |
| Ugi                 | 20260827 | 2,257             | 3,072    | 73.47     |
| Aza-Michael         | 20260825 | 2,351             | 3,072    | 76.53     |
| Aza-Michael         | 20260826 | 2,504             | 3,072    | 81.51     |
| Aza-Michael         | 20260827 | 2,284             | 3,072    | 74.35     |
| Reductive amination | 20260825 | 1,186             | 3,072    | 38.61     |
| Reductive amination | 20260826 | 454               | 3,072    | 14.78     |
| Reductive amination | 20260827 | 1,242             | 3,072    | 40.43     |

#### C.4 REACTION SEMANTICS ALTER GENERATION RATHER THAN MERELY FILTERING ITS OUTPUT

Under matched attempt budgets, conditioned generation produced 786.8, 774.6 and 312.7 verified exact-L1 products per 1,000 attempts for Ugi, repeated aza-Michael and repeated reductive amination, compared with 114.4, 39.7 and 131.7 for matched null generation followed by post-hoc filtering (Table 4). The paired conditioned-minus-post-hoc differences were 67.24 percentage points for Ugi (three-seed range 60.84 to 79.95), 73.49 for aza-Michael (69.69 to 78.45), and 18.10 for reductive amination (2.51 to 26.04).

The cyclic control retained precursor roles, reaction-core coordinates and program depth but assigned the wrong reaction identity. Relative to correct conditioning, this intervention changed verified exact-L1 yield by 38.43 percentage points for Ugi (three-seed range 32.00 to 49.41), 73.67 for aza-Michael (71.13 to 77.02), and 13.15 for reductive amination (-3.91 to 22.01). Every returned exact trace passed replay by verifier construction.

A separate inference-time intervention held the trained network and Ugi request fixed while changing only program coordinates. The factual verified exact-L1 yield was 39.69%. Nulling all program coordinates reduced it to 3.62%, a 36.07-point loss (paired differences 37.24, 34.28, 36.69, spanning 34.28 to 37.24). Reversing the role assignment caused a 4.49-point loss (paired range 1.37 to 8.27), and substituting the reductive-amination program token caused a 22.60-point loss (paired range 14.06 to 27.47). The aza-Michael-token contrast was unresolved at 2.00 points (paired range -2.18 to 5.60). Thus the coordinates affect generation, but the evidence does not support an equal or individually identifiable effect for every coordinate.

Table 6: **Method-blind structural-realism diagnostic.** Every entry is mean  $\pm$  sample standard deviation across three training seeds and retains all 3,072 requested attempts in the denominator. FP and descriptor P/C report manifold precision per 1,000 attempts and coverage of the source-study-held-out R0 reference in percent. C2ST is grouped classifier two-sample area under the receiver operating characteristic curve; 0.5 is indistinguishable.  $N_{\text{eff}}$  is the Shannon effective molecule count. N/E denotes a metric that cannot be estimated. This post-hoc structural diagnostic is not a property, activity or synthesis-success score.

| Method            | Connected/1k     | FP P/C                        | Desc. P/C                      | C2ST              | $N_{\text{eff}}$   | Diversity         |
|-------------------|------------------|-------------------------------|--------------------------------|-------------------|--------------------|-------------------|
| RGFN              | 0.0 $\pm$ 0.0    | 0.0 $\pm$ 0.0/0.00 $\pm$ 0.00 | 0.0 $\pm$ 0.0/0.00 $\pm$ 0.00  | N/E               | 0.0 $\pm$ 0.0      | N/E               |
| DeFoG             | 346.5 $\pm$ 42.3 | 0.0 $\pm$ 0.0/0.00 $\pm$ 0.00 | 17.3 $\pm$ 1.4/1.45 $\pm$ 0.06 | 0.961 $\pm$ 0.007 | 1064.3 $\pm$ 129.9 | 0.726 $\pm$ 0.013 |
| unconditional     |                  |                               |                                |                   |                    |                   |
| GenMol/SAFE       | 627.5 $\pm$ 6.2  | 0.0 $\pm$ 0.0/0.00 $\pm$ 0.00 | 0.0 $\pm$ 0.0/0.00 $\pm$ 0.00  | N/E               | 3.2 $\pm$ 0.1      | 0.873 $\pm$ 0.009 |
| Finite catalogue  | 1000.0 $\pm$ 0.0 | 0.2 $\pm$ 0.2/0.05 $\pm$ 0.05 | 54.6 $\pm$ 4.4/2.02 $\pm$ 0.17 | 0.971 $\pm$ 0.007 | 2395.5 $\pm$ 31.7  | 0.720 $\pm$ 0.000 |
| oracle            |                  |                               |                                |                   |                    |                   |
| Learned inventory | 1000.0 $\pm$ 0.0 | 0.1 $\pm$ 0.2/0.02 $\pm$ 0.03 | 56.6 $\pm$ 7.2/1.97 $\pm$ 0.46 | 0.974 $\pm$ 0.002 | 2443.0 $\pm$ 55.8  | 0.718 $\pm$ 0.001 |
| selector          |                  |                               |                                |                   |                    |                   |
| FORGE             | 866.0 $\pm$ 90.2 | 0.0 $\pm$ 0.0/0.00 $\pm$ 0.00 | 8.8 $\pm$ 2.3/1.16 $\pm$ 0.20  | 0.973 $\pm$ 0.019 | 2479.4 $\pm$ 177.2 | 0.621 $\pm$ 0.013 |
| Transformer       |                  |                               |                                |                   |                    |                   |

### C.5 WHOLE-GRAPH GENERATION RECOVERS PRECURSOR IDENTITIES ABSENT FROM THE TRAINING CATALOGUE

Table 2 defines the common Ugi benchmark used to compare FORGE with external synthesis-aware generators, matched internal controls, and the finite-catalogue oracle. Every yield is normalized by the number of generation attempts, including attempts that fail before producing a valid molecule. All included methods were run under the common split, attempt budget, seed set and verifier.

The learned finite-inventory selector achieved 1000.0  $\pm$  0.0 verified exact-L1 products per 1,000 attempts, but its finite action space imposed zero verifier-recovered component-novel yield. FORGE produced 786.8  $\pm$  100.8 verified exact-L1 products and 751.2  $\pm$  80.2 distinct verified exact-L1 products per 1,000 attempts. Among whole-molecule external generators, unconditional DeFoG produced 280.6  $\pm$  26.6 verified exact-L1 products per 1,000 attempts and recovered held components at 15.5  $\pm$  2.4 per 1,000 attempts, compared with 0.1  $\pm$  0.2 for FORGE. GenMol/SAFE produced 627.5  $\pm$  6.2 valid connected outputs but no exact Ugi-L1 product, while RGFN emitted no admissible molecular output under the strict native contract. These outcomes expose separate tradeoffs between reaction consistency, verifier-recovered component novelty and recovery of specifically reserved components. They do not establish product-level strong catalogue escape.

No evaluated method reproduced the full held-out observed-lipid distribution: where estimable, grouped C2ST AUC remained between 0.961 and 0.974, and fingerprint-manifold precision was zero or near zero (Table 6). GenMol/SAFE’s high pairwise diversity was paired with an effective count of only 3.2  $\pm$  0.1 molecules, confirming severe frequency collapse. FORGE retained 2479.4  $\pm$  177.2 effective molecules, but only 8.8  $\pm$  2.3 attempts per 1,000 entered the descriptor manifold. We therefore interpret this analysis as a multi-axis structural diagnostic rather than a single measure of whether a generated lipid is reasonable.

### C.6 JOINT WHOLE-LIPID DENOISING HAS REACTION-DEPENDENT ADVANTAGES OVER ROLE ISOLATION

The full Transformer produced 791.8 Ugi and 762.5 repeated aza-Michael exact-L1 products per 1,000 attempts, compared with 52.0 and 393.7 for the parameter-matched role-isolated FACT model (Table 7). The full Transformer and FACT-matched model each had 5.33 million trainable parameters; FACT-generous had 29.40 million. The higher-capacity FACT control improved to 296.7 for Ugi and 687.0 for repeated aza-Michael addition, but remained below the full Transformer in every seed for both programs. The direction reversed for repeated reductive amination: the full Transformer produced 318.1 exact-L1 products per 1,000 attempts, whereas FACT-matched and FACT-generous produced 509.9 and 561.3. Thus, joint whole-lipid denoising improved exact assembly for Ugi and repeated aza-Michael addition but was not uniformly superior across reaction programs. Because the prespecified

Table 7: **Transformer architecture and mechanism ablations.** The five Transformer intervention rows change only the named mechanism from the full model; the two FACT rows instead generate precursor roles independently at matched or increased capacity. All arms use identical data folds, corruption, context schedule, checkpoint rule and sampling budget. Entries report the mean  $\pm$  sample standard deviation across three independent training seeds. Held-out loss is the mean denoising loss at  $t = 0.5$  across the three component-disjoint program partitions. Cross-role fidelity is the Frobenius distance to the held-out residualized descriptor matrix, for which lower values indicate closer agreement. Exact-L1 yields use all 3,072 attempts per program and seed as the denominator.

| Arm                          | Held-out loss | Ugi L1/1k         | BL L1/1k          | LX L1/1k          | Held-comp. L1/1k | Cross-role fidelity | Diversity     |
|------------------------------|---------------|-------------------|-------------------|-------------------|------------------|---------------------|---------------|
| Input-only program           | 8.5 $\pm$ 0.7 | 737.3 $\pm$ 107.3 | 779.8 $\pm$ 35.1  | 277.3 $\pm$ 56.2  | 0.1 $\pm$ 0.2    | 1.8 $\pm$ 0.1       | 0.6 $\pm$ 0.0 |
| No role loss                 | 7.9 $\pm$ 0.1 | 771.6 $\pm$ 109.5 | 772.1 $\pm$ 42.2  | 289.3 $\pm$ 152.0 | 0.1 $\pm$ 0.2    | 1.9 $\pm$ 0.0       | 0.6 $\pm$ 0.0 |
| No core loss                 | 7.9 $\pm$ 0.1 | 753.7 $\pm$ 132.4 | 772.8 $\pm$ 46.2  | 300.5 $\pm$ 133.3 | 0.1 $\pm$ 0.2    | 1.9 $\pm$ 0.1       | 0.6 $\pm$ 0.0 |
| No routed adapters           | 7.8 $\pm$ 0.3 | 718.6 $\pm$ 143.9 | 779.1 $\pm$ 40.7  | 326.7 $\pm$ 59.3  | 0.0 $\pm$ 0.0    | 1.9 $\pm$ 0.2       | 0.6 $\pm$ 0.0 |
| No gradient conflict control | 8.0 $\pm$ 0.2 | 798.1 $\pm$ 102.4 | 768.9 $\pm$ 43.2  | 326.9 $\pm$ 154.5 | 0.1 $\pm$ 0.2    | 2.0 $\pm$ 0.1       | 0.6 $\pm$ 0.0 |
| FACT-matched                 | 7.1 $\pm$ 0.2 | 52.0 $\pm$ 15.6   | 393.7 $\pm$ 66.7  | 509.9 $\pm$ 52.7  | 0.0 $\pm$ 0.0    | 1.4 $\pm$ 0.3       | 0.6 $\pm$ 0.0 |
| FACT-generous                | 8.2 $\pm$ 0.3 | 296.7 $\pm$ 186.0 | 687.0 $\pm$ 115.1 | 561.3 $\pm$ 72.2  | 0.1 $\pm$ 0.2    | 1.2 $\pm$ 0.0       | 0.6 $\pm$ 0.0 |
| FORGE Transformer            | 7.9 $\pm$ 0.2 | 791.8 $\pm$ 108.2 | 762.5 $\pm$ 46.9  | 318.1 $\pm$ 159.1 | 0.1 $\pm$ 0.2    | 1.9 $\pm$ 0.2       | 0.6 $\pm$ 0.0 |

conditional-dependence diagnostic was negative, this architecture contrast does not establish detectable cross-role dependence in the present data.

Removing layerwise program cross-attention reduced mean exact-L1 yield by 54.5 products per 1,000 attempts for Ugi and 40.8 for repeated reductive amination, while increasing repeated aza-Michael yield by 17.4. Removing the role or core objective and removing routed adapters also produced program-dependent changes rather than a uniform loss. The arm without gradient-conflict control slightly exceeded the full Transformer mean in all three programs. We therefore do not claim that every introduced module independently improves exact-L1 yield. These one-at-a-time interventions identify which effects replicate across chemistries; they are not pooled into a composite architecture score.

At the matched sampling budget, FORGE produced 729.3 verifier-recovered component-novel Ugi products per 1,000 attempts, averaged across the three independent seeds. The corresponding values were 313.9 for aza-Michael addition and 34.8 for reductive amination. The finite-component assembler produced zero by construction because every sampled product uses only components from the training catalogue (Table 8). The catalogue achieved exact-L1 yields of 1,000.0, 723.2 and 1,000.0 products per 1,000 attempts, compared with 786.8, 774.6 and 312.7 for FORGE. The corresponding raw validities were 1,000.0, 723.2 and 1,000.0 for the catalogue and 866.0, 774.6 and 365.0 for FORGE. We report these quantities separately because reaction validity and verifier-relative component novelty measure different capabilities. Direct membership in the complete catalogue-reachable product set was not computed, so these results are not reported as strong catalogue escape.

In a distinct-product Ugi role audit, 19,520 of 26,235 products (74.4%) contained at least one verifier-returned component constitution absent from the training registry. Across all Ugi products admitted to the novelty audit, 21,960 of 26,235 distinct products (83.7%) were absent from the full 112,386-product constitutional reference corpus. FORGE effective component counts were 320.8, 156.9 and 102.9, compared with 86.4, 23.4 and 16.9 for the finite catalogue. The Ugi catalogue contained 185 amines, 75 aldehydes and 35 isocyanides and covered 100% of training products but 0% of the calibration and component-disjoint held-out products. Its Cartesian tuple ceilings were 485,625 for Ugi, 774 for aza-Michael and 490 for reductive amination; these count possible component-program inputs rather than unique forward products. The exact finite catalogue has high transform consistency, whereas whole-graph generation can produce verified exact-L1 products with verifier-recovered precursor identities absent from that inventory.

Table 8: **Whole-graph generation versus a finite-component assembler.** Verified exact-L1, distinct exact-L1 and verifier-recovered component-novel yields are reported per 1,000 generation attempts. A component-novel product has exactly one verifier-returned exact-L1 decomposition and contains at least one component constitution absent from the training catalogue. This verifier-relative metric is not strong catalogue escape because complete catalogue-product reachability was not computed. The catalogue has zero input-component novelty by construction, but may achieve higher raw validity or exact-L1 yield. FORGE-minus-catalogue contrasts use the three seeds as descriptive paired units; no three-seed confidence interval is inferred.

|                                       | Ugi         |             | Aza-Michael |             | Reductive amination |             |
|---------------------------------------|-------------|-------------|-------------|-------------|---------------------|-------------|
|                                       | FORGE       | Finite cat. | FORGE       | Finite cat. | FORGE               | Finite cat. |
| Valid/1k                              | 866.0       | 1000.0      | 774.6       | 723.2       | 365.0               | 1000.0      |
| Verified exact L1/1k                  | 786.8       | 1000.0      | 774.6       | 723.2       | 312.7               | 1000.0      |
| Distinct exact L1/1k                  | 751.2       | 837.6       | 738.0       | 147.5       | 279.7               | 52.1        |
| Verifier-recovered component-novel/1k | 729.3       | 0.0         | 313.9       | 0.0         | 34.8                | 0.0         |
| Whole novelty                         | 99.8        | 16.9        | 99.7        | 39.8        | 98.1                | 44.8        |
| Component novelty                     | 96.2        | 0.0         | 99.2        | 0.0         | 100.0               | 0.0         |
| Diversity/ $N_{\text{eff}}$           | 0.621/320.8 | 0.719/86.4  | 0.521/156.9 | 0.402/23.4  | 0.624/102.9         | 0.514/16.9  |

### C.7 NATIVE RECOVERY OF HELD COMPONENTS REMAINS WEAK

The component-disjoint Ugi fold was never used for training, checkpoint selection or inventory construction. Under the common native-generation assessment, the full Transformer recovered only 1 exact-L1 product containing a designated held component in 9,216 attempts, or 0.11 products per 1,000 attempts averaged across three seeds. This is a negative generalization result. It does not conflict with the broader verifier-relative novelty measurements above: generating a component absent from the training registry is easier than recovering one of the specifically reserved held-component identities. We therefore do not claim strong held-component recovery, role-specific held-component coverage, or zero-shot held-reaction-family generalization.

### C.8 RECURSIVE SYNTHESIS PROGRAMS CONNECT GENERATED LIPIDS TO EVIDENCE-QUALIFIED MATERIALS

Table 9 separates a method’s native identity source and route representation from evidence obtained under the common FORGE assessor. We freeze the union of exact-L1 components from all methods and seeds before route assessment, hide method membership from the evidence pass, and require one complete, unresolved or search-censored disposition for every component before unsealing membership. Emitting a reaction sequence is not counted as a complete dossier unless the final assembly and every upstream step pass exact forward verification and every terminal material closes under the frozen evidence policy. We do not report a separate AiZynthFinder row: a proposal-only route is a hypothesis and cannot change exact evidence closure until every step passes the same forward-verification and terminal-evidence gates (Genheden et al., 2020).

The frozen union contained 11,313 distinct role-constitution components, but the exact-source library closed only 29. Consequently, FORGE produced  $786.8 \pm 100.8$  verified exact-L1 products per 1,000 attempts, yet  $0.0 \pm 0.0$  complete dossiers and  $786.8 \pm 100.8$  abstentions under this library. The learned inventory selector closed  $0.7 \pm 0.3$  dossiers per 1,000 attempts through occasional reuse of covered training components. This is a negative result about current exact evidence coverage, not a comparison of intrinsic synthesizability: components outside the frozen library were not searched and remain unknown.

The completed bounded route cascade starts from a frozen, route-blinded shortlist of 256 distinct Ugi products rather than from the full production-attempt population. All 256 products passed exact L1 reverification; 255 entered route assessment and one silicon-containing state was support-censored. Across the full shortlist, 19 closed under the exact-evidence and terminal-material policy, 236 remained unresolved and 1 was search-censored. The products contained 158 unique component targets: 36 complete, 120 unresolved and 2 search-censored. The proposal engines added graph-consistent hypotheses but closed no additional

Table 9: **Common synthesis-evidence assessment of Ugi generative methods.** Descriptive support fields record what each method is permitted to use at generation time. Numerical evidence fields are populated only after applying the same method-blind union of exact-L1 components, upstream forward-verification, terminal-evidence, and abstention policy to each method’s outputs. N/P means that the method does not produce that object; it is not interpreted as a measured zero. All numerical fields are reported per 1,000 generation attempts as mean  $\pm$  sample standard deviation across three seeds. An abstention records missing exact evidence, not a failed synthesis; the frozen exact-source component library is deliberately non-exhaustive.

| Method                     | Molecular identity source                           | Finite vocab. | Verified L1/1k    | Verified upstream | Terminal evidence | Complete dossier | Abstention        |
|----------------------------|-----------------------------------------------------|---------------|-------------------|-------------------|-------------------|------------------|-------------------|
| RGFN                       | reaction $\times$ building block                    | yes           | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0    | 0.0 $\pm$ 0.0     |
| DeFoG unconditional        | generated atoms and bonds                           | no            | 280.6 $\pm$ 26.6  | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0    | 280.6 $\pm$ 26.6  |
| GenMol/SAFE                | generated fragment sequence                         | no            | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0    | 0.0 $\pm$ 0.0     |
| Finite catalogue oracle    | train components                                    | yes           | 1000.0 $\pm$ 0.0  | 0.1 $\pm$ 0.2     | 0.1 $\pm$ 0.2     | 0.1 $\pm$ 0.2    | 999.9 $\pm$ 0.2   |
| Learned inventory selector | selected train components                           | yes           | 1000.0 $\pm$ 0.0  | 0.7 $\pm$ 0.3     | 0.7 $\pm$ 0.3     | 0.7 $\pm$ 0.3    | 999.3 $\pm$ 0.3   |
| Shared-null + post-hoc     | generated atoms and bonds                           | no            | 114.4 $\pm$ 10.9  | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0    | 114.4 $\pm$ 10.9  |
| FACT-matched               | role-isolated denoising on shared count-only layout | no            | 52.0 $\pm$ 15.6   | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0    | 52.0 $\pm$ 15.6   |
| FORGE Transformer          | generated atoms and bonds                           | no            | 786.8 $\pm$ 100.8 | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0     | 0.0 $\pm$ 0.0    | 786.8 $\pm$ 100.8 |

Table 10: **Predicted HeLa messenger-transfection potential under matched generation budgets.** The support-enriched arm uses the already promoted applicability proposal; the nested arm reallocates morphology mass using the frozen weak potency signal. Conservative-high counts are computational predictions, not transfection measurements. The comparison uses the same generator-call and oracle budgets.

| Arm                     | Attempts | Verified L1 | Eligible before budget | Oracle budget | Unique conservative-high |
|-------------------------|----------|-------------|------------------------|---------------|--------------------------|
| Support-enriched        | 3,072    | 2955        | 90                     | 38            | 27                       |
| Nested potency proposal | 3,072    | 2962        | 82                     | 38            | 23                       |

exact routes. Most unresolved components were absent from the bounded evidence index and were never expanded, so the result measures closure under this frozen cascade rather than molecular synthesizability.

### C.9 GUIDANCE DIAGNOSTICS DEFINE THE LIMITS OF ADDITIONAL OPTIMIZATION

At synthesis-guidance strength 0.25, the matched controller evaluated 64 terminal particles and retained 56 canonical representatives per arm. It changed only 8 terminal identities and returned 2 unique route-ready products, compared with 3 after post-hoc assessment. Only five of 48 four-particle morphology groups contained mixed route-readiness values, leaving most ancestry decisions uniform. This negative result does not support promoting mid-trajectory synthesis tilting. The production method therefore remains complete generation followed by proposal-augmented L1/L2/L3 route assessment.

We evaluate biological guidance only in the bounded HeLa messenger-transfection-potential (mTP) diagnostic. A group-disjoint preterminal classifier detected weak morphology-level signal, with ROC AUC 0.623 (95% interval 0.565 to 0.679) and a 1.177-fold observed top-quartile gain. We then compared a frozen applicability-enriched morphology proposal with a nested potency-enriched proposal using 3,072 generator calls per arm. Table 10 reports the matched terminal outcome.

The nested-minus-support difference was -1.30 conservative-high products per 1,000 generator calls (95% interval -2.28 to 0.98). The promotion gate failed. These computations evaluate predictor behavior within a declared applicability domain; they do not establish improved biological activity, delivery or experimental potency.

## D EXTENDED COMPUTATIONAL REPORTING TABLES

The following tables define the complete numerical reporting contract for the common Ugi benchmark. Completed matched controls, the learned inventory selector and three external generators are populated from hash-pinned three-seed results. The method-blind L2/L3 evidence assessment is reported in Table 9. Compute accounting remains in the generated evidence receipt; unavailable training-time fields preclude a cross-method compute-matching claim.

Table 11: **Seed-level common Ugi benchmark results.** Rows are expanded once a method has a hash-verified three-seed result vector. The main table reports the corresponding mean  $\pm$  sample standard deviation; generated molecules are not treated as independent training replicates.

| Method                     | Seed(s)  | Valid/1k | Verified L1/1k | Distinct L1/1k | Comp.-novel/1k | Held-comp. L1/1k | Diversity |
|----------------------------|----------|----------|----------------|----------------|----------------|------------------|-----------|
| RGFN                       | 20260825 | 0.0      | 0.0            | 0.0            | 0.0            | 0.0              | N/E       |
| RGFN                       | 20260826 | 0.0      | 0.0            | 0.0            | 0.0            | 0.0              | N/E       |
| RGFN                       | 20260827 | 0.0      | 0.0            | 0.0            | 0.0            | 0.0              | N/E       |
| DeFoG unconditional        | 20260825 | 299.2    | 251.0          | 251.0          | 251.0          | 13.7             | 0.711     |
| DeFoG unconditional        | 20260826 | 359.7    | 288.4          | 288.4          | 288.4          | 18.2             | 0.733     |
| DeFoG unconditional        | 20260827 | 380.5    | 302.4          | 302.4          | 302.4          | 14.6             | 0.736     |
| GenMol/SAFE                | 20260825 | 627.3    | 0.0            | 0.0            | 0.0            | 0.0              | 0.883     |
| GenMol/SAFE                | 20260826 | 633.8    | 0.0            | 0.0            | 0.0            | 0.0              | 0.869     |
| GenMol/SAFE                | 20260827 | 621.4    | 0.0            | 0.0            | 0.0            | 0.0              | 0.866     |
| Learned inventory selector | 20260825 | 1000.0   | 1000.0         | 863.0          | 0.0            | 0.0              | 0.717     |
| Learned inventory selector | 20260826 | 1000.0   | 1000.0         | 835.6          | 0.0            | 0.0              | 0.717     |
| Learned inventory selector | 20260827 | 1000.0   | 1000.0         | 849.9          | 0.0            | 0.0              | 0.715     |
| Finite catalogue oracle    | 20260825 | 1000.0   | 1000.0         | 841.5          | 0.0            | 0.0              | 0.719     |
| Finite catalogue oracle    | 20260826 | 1000.0   | 1000.0         | 843.1          | 0.0            | 0.0              | 0.719     |
| Finite catalogue oracle    | 20260827 | 1000.0   | 1000.0         | 828.1          | 0.0            | 0.0              | 0.719     |
| Shared-null + post-hoc     | 20260825 | 183.9    | 114.3          | 109.7          | 109.7          | 0.3              | 0.738     |
| Shared-null + post-hoc     | 20260826 | 187.5    | 103.5          | 98.3           | 98.3           | 0.7              | 0.763     |
| Shared-null + post-hoc     | 20260827 | 212.6    | 125.3          | 117.5          | 117.5          | 0.3              | 0.761     |
| FACT-matched               | 20260825 | 59.2     | 49.5           | 48.8           | 48.8           | 0.0              | 0.676     |
| FACT-matched               | 20260826 | 81.7     | 68.7           | 66.1           | 66.1           | 0.0              | 0.658     |
| FACT-matched               | 20260827 | 41.3     | 37.8           | 37.4           | 37.4           | 0.0              | 0.668     |
| FACT-generous              | 20260825 | 391.0    | 161.8          | 161.1          | 161.1          | 0.3              | 0.782     |
| FACT-generous              | 20260826 | 747.1    | 508.8          | 488.3          | 488.3          | 0.0              | 0.768     |
| FACT-generous              | 20260827 | 363.9    | 219.4          | 215.5          | 215.5          | 0.0              | 0.771     |
| FORGE Transformer          | 20260825 | 801.4    | 722.7          | 702.1          | 702.1          | 0.0              | 0.617     |
| FORGE Transformer          | 20260826 | 969.1    | 903.0          | 843.8          | 843.8          | 0.3              | 0.613     |
| FORGE Transformer          | 20260827 | 827.5    | 734.7          | 707.7          | 707.7          | 0.0              | 0.634     |

Table 12: **Reaction decomposition outcomes under the common Ugi verifier.** Verified decomposition coverage is conditioned on molecular validity. Replay of every tuple in the returned exact set is a verifier invariant, not an estimate of reverse-search precision. Abstention and exact ambiguity retain their prespecified denominators. N/E denotes a conditional metric with no eligible valid product or returned decomposition.

| Method                     | Valid products     | Verified coverage | Replay invariant | Abstention      | Exact ambiguity |
|----------------------------|--------------------|-------------------|------------------|-----------------|-----------------|
| RGFN                       | 0.0 $\pm$ 0.0      | N/E               | N/E              | N/E             | N/E             |
| DeFoG unconditional        | 1064.3 $\pm$ 129.9 | 81.2 $\pm$ 2.4    | 100.0 $\pm$ 0.0  | 18.8 $\pm$ 2.4  | 0.0 $\pm$ 0.0   |
| GenMol/SAFE                | 1927.7 $\pm$ 19.0  | 0.0 $\pm$ 0.0     | N/E              | 100.0 $\pm$ 0.0 | 0.0 $\pm$ 0.0   |
| Learned inventory selector | 3072.0 $\pm$ 0.0   | 100.0 $\pm$ 0.0   | 100.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| Finite catalogue oracle    | 3072.0 $\pm$ 0.0   | 100.0 $\pm$ 0.0   | 100.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| Shared-null + post-hoc     | 598.0 $\pm$ 47.9   | 58.8 $\pm$ 3.5    | 100.0 $\pm$ 0.0  | 41.2 $\pm$ 3.5  | 0.0 $\pm$ 0.0   |
| FACT-matched               | 186.7 $\pm$ 62.1   | 86.3 $\pm$ 4.4    | 100.0 $\pm$ 0.0  | 13.7 $\pm$ 4.4  | 0.0 $\pm$ 0.0   |
| FACT-generous              | 1538.0 $\pm$ 656.9 | 56.6 $\pm$ 13.7   | 100.0 $\pm$ 0.0  | 43.4 $\pm$ 13.7 | 0.0 $\pm$ 0.0   |
| FORGE Transformer          | 2660.3 $\pm$ 277.1 | 90.7 $\pm$ 2.2    | 100.0 $\pm$ 0.0  | 9.3 $\pm$ 2.2   | 0.0 $\pm$ 0.0   |

## E THEORETICAL PROPERTIES

This appendix establishes the formal properties of FORGE. We prove correctness and bounded completeness of the sparse representation, characterize finite-component reachability, prove fixed-coordinate invariance and Bayes optimality of the primary coordinatewise denoising loss, connect those posterior marginals to the local continuous-time rate, prove ideal population recovery of the terminal program-conditioned graph law, quantify matched-budget post-hoc rejection, and state termination plus the fail-closed route implementation invariant.

These results concern representation, population objectives, and computational verification. They do not imply that finite-data training assigns positive probability to every representable graph, that registered transformations cover arbitrary chemistry, that bounded route search is complete, or that a returned dossier guarantees experimental synthesis success.

### E.1 NOTATION AND SETUP

We use the following notation consistently.

| Symbol                                       | Meaning                                            |
|----------------------------------------------|----------------------------------------------------|
| $\Sigma_{\text{atom}}, \Sigma_{\text{bond}}$ | atom-state and bond-state alphabets                |
| $N, K_{\text{cl}}$                           | maximum active nodes and closure-edge budget       |
| $L_{\text{samp}}$                            | finite Euler-step count                            |
| $\mathcal{I}$                                | six categorical state families                     |
| $\mathcal{F}(P), \mathcal{V}(P)$             | fixed and variable coordinate sets                 |
| $\mathfrak{C}_P$                             | admissible precursor-program inputs                |
| $\mathfrak{B}_{\text{route}}$                | frozen route-assessment contract                   |
| $\mathfrak{D}, \mathcal{D}_{\text{out}}$     | dossier space and attempt-indexed dossier sequence |
| $Q$                                          | proposal law in the post-hoc comparison            |

A molecular graph is  $G = (V, E, \ell_V, \ell_E)$ , where the underlying graph is finite, connected, undirected and simple,  $\ell_V : V \rightarrow \Sigma_{\text{atom}}$ , and  $\ell_E : E \rightarrow \Sigma_{\text{bond}}$ . Bond multiplicity belongs to  $\ell_E$ ; parallel edges are not permitted. Define the set of label-preserving isomorphism classes

$$\mathfrak{G}_{N, K_{\text{cl}}} = \{[G]_{\cong} : |V| \leq N, \beta(G) = |E| - |V| + 1 \leq K_{\text{cl}}, \ell_V(V) \subseteq \Sigma_{\text{atom}}, \ell_E(E) \subseteq \Sigma_{\text{bond}}\}. \quad (42)$$

Let  $\bar{\mathfrak{X}}_{N, K_{\text{cl}}}$  be the full padded categorical product space. A state belongs to its well-formed subset  $\mathfrak{X}_{N, K_{\text{cl}}}$  exactly when:

1. for some  $1 \leq n \leq N$ , the active nodes are exactly  $\{1, \dots, n\}$ , every active node has a label in  $\Sigma_{\text{atom}}$ , and inactive node slots carry the null state;
2. node 1 is the unique root and carries the null parent-edge sentinel;
3. every  $i > 1$  has  $\pi_i \in \{1, \dots, i-1\}$  and a parent-edge label in  $\Sigma_{\text{bond}}$ ;
4. every active closure slot joins two distinct endpoints in  $\{1, \dots, n\}$ , carries a label in  $\Sigma_{\text{bond}}$ , and duplicates neither a tree edge nor another closure edge;
5. active closure triples are ordered lexicographically by endpoint pair; and
6. at most  $K_{\text{cl}}$  closure slots are active and every unused slot carries designated null endpoint and bond states.

The raw decoder returns a typed failure on every malformed state:

$$\text{Dec} : \bar{\mathfrak{X}}_{N, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (43)$$

## E.2 SPARSE GRAPH REPRESENTATION

**Lemma E.1** (Tree residual count). *Let  $G = (V, E)$  be a connected finite graph and let  $T$  be any spanning tree of  $G$ . Then*

$$|E \setminus E(T)| = |E| - |V| + 1 = \beta(G). \quad (44)$$

*Proof.* A spanning tree on  $|V|$  vertices contains exactly  $|V| - 1$  edges. Because  $E(T) \subseteq E$ ,

$$|E \setminus E(T)| = |E| - |E(T)| = |E| - (|V| - 1) = |E| - |V| + 1. \quad (45)$$

The final expression is the cyclomatic number of a connected graph.  $\square$

**Theorem E.2** (Correctness and bounded completeness of the sparse representation). *The decoder*

$$\text{Dec} : \mathfrak{X}_{N, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N, K_{\text{cl}}} \quad (46)$$

*is well defined and surjective up to label-preserving graph isomorphism:*

1. *every  $X \in \mathfrak{X}_{N, K_{\text{cl}}}$  decodes to a graph in  $\mathfrak{G}_{N, K_{\text{cl}}}$ ; and*
2. *for every  $G \in \mathfrak{G}_{N, K_{\text{cl}}}$ , there exists  $X \in \mathfrak{X}_{N, K_{\text{cl}}}$  such that  $\text{Dec}(X) \cong G$ .*

*Proof.* Let  $G = (V, E, \ell_V, \ell_E) \in \mathfrak{G}_{N, K_{\text{cl}}}$ , and choose a spanning tree  $T = (V, E_T)$  with  $E_T \subseteq E$ . Choose a root of  $T$  and enumerate its vertices  $v_1, \dots, v_n$  in any root-first traversal. Every non-root vertex then appears after its parent.

Assign node  $i$  the label of  $v_i$ . For each  $i > 1$ , let  $\pi_i < i$  be the index of the parent of  $v_i$  in  $T$ , and assign the corresponding tree-edge label. By Lemma E.1, the residual edge set  $E \setminus E_T$  contains  $\beta(G) \leq K_{\text{cl}}$  edges. Orient each residual edge by increasing endpoint index, attach its original edge label, and sort the resulting triples lexicographically by endpoint pair. The resulting state is an element of  $\mathfrak{X}_{N, K_{\text{cl}}}$ .

The decoder reconstructs every edge in  $E_T$  from the parent pointers and every edge in  $E \setminus E_T$  from the closure triples. It also preserves all node and edge labels. The map  $i \mapsto v_i$  is therefore a label-preserving isomorphism from  $\text{Dec}(X)$  to  $G$ .

Conversely, let  $X \in \mathfrak{X}_{N, K_{\text{cl}}}$  have  $n$  active nodes. Since  $\pi_i < i$ , repeatedly following parent pointers from any node strictly decreases its index and must terminate at the root. The  $n - 1$  parent edges therefore form a connected acyclic graph, hence a spanning tree. If the state contains  $k \leq K_{\text{cl}}$  closure edges, the decoded graph has

$$|E| = (n - 1) + k \quad (47)$$

distinct edges. Its cyclomatic number is consequently

$$|E| - |V| + 1 = (n - 1 + k) - n + 1 = k \leq K_{\text{cl}}. \quad (48)$$

The remaining well-formedness conditions preserve the declared node and edge vocabularies and exclude loops or duplicate edges. Thus  $\text{Dec}(X) \in \mathfrak{G}_{N, K_{\text{cl}}}$ .  $\square$

Global completeness does not by itself imply compatibility with a program face  $\Omega_P$ . FORGE therefore restricts the morphology prior to

$$\mathcal{P}_{\text{adm}} = \{P : \mathfrak{X}_{N, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\} \quad (49)$$

and treats  $\text{Enc}_P(G, \alpha) \in \mathfrak{X}_{N, K_{\text{cl}}} \cap \Omega_P$  as an adapter-level implementation invariant checked for every admitted corpus record.

## E.3 FINITE-COMPONENT REACHABILITY

**Proposition E.3** (Finite-component reachability). *Assume that  $\mathfrak{C}_P(\mathcal{C})$  is finite and that every forward-outcome set is finite. Then*

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq \sum_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|. \quad (50)$$

If  $\mathfrak{C}_P(\mathcal{C}) \neq \emptyset$  and

$$m_{\max, P}(\mathcal{C}) = \max_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|, \quad (51)$$

then

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq m_{\max, P}(\mathcal{C}) |\mathfrak{C}_P(\mathcal{C})|. \quad (52)$$

Moreover,

$$G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C}) \iff \mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C}) \neq \emptyset. \quad (53)$$

*Proof.* The reachable set is a finite union of forward-outcome sets. The cardinality of a union is at most the sum of the cardinalities of its members, which proves the first inequality. Bounding each summand by  $m_{\max, P}(\mathcal{C})$  proves the second.

By definition,  $G$  belongs to the union precisely when at least one  $\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})$  satisfies  $G \in \mathcal{F}_P(\mathbf{C})$ . This is equivalent to membership in  $\mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C})$ .  $\square$

**Corollary E.4** (Catalogue assemblers cannot strongly escape). *For every  $G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C})$ ,  $\text{Escape}^*(G; P, \mathcal{C}) = 0$ . Consequently any assembler whose outputs are contained in  $\mathcal{G}_{\text{cat}}(P, \mathcal{C})$  has zero strong-catalogue-escape yield per attempt.*

*Proof.* Membership in  $\mathcal{G}_{\text{cat}}(P, \mathcal{C})$  supplies a catalogue tuple in  $\mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C})$ , contradicting the second conjunct in the definition of strong escape.  $\square$

#### E.4 PROGRAM-MASKED DISCRETE FLOW

For a program  $P$ , let  $\mathcal{I}(P)$  index all categorical coordinates, with coordinate alphabet  $\mathcal{X}_i$ . Partition it into fixed coordinates  $\mathcal{F}(P)$  and variable coordinates  $\mathcal{V}(P)$ . Let  $x_{P,i}$  be the prescribed value for  $i \in \mathcal{F}(P)$  and define

$$\Omega_P = \{x : x_i = x_{P,i} \text{ for every } i \in \mathcal{F}(P)\}. \quad (54)$$

The projection  $\Pi_P$  restores  $x_{P,i}$  on  $\mathcal{F}(P)$  and is the identity on  $\mathcal{V}(P)$ .

The program-compatible source factorizes as

$$q_0^P(x) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{P,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} q_{0,i}^P(x_i), \quad q_{0,i}^P(a) > 0 \quad \forall a \in \mathcal{X}_i. \quad (55)$$

For a clean endpoint  $x_1 \in \Omega_P$ , the independent-coordinate conditional path is

$$q_{t|1}^P(x | x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{1,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1-t)q_{0,i}^P(x_i) + t\mathbf{1}\{x_i = x_{1,i}\}]. \quad (56)$$

Thus each coordinate is linearly interpolated conditional on the endpoint, while the joint law is their product. It is not the joint mixture  $(1-t)q_0^P + tq_1^P$ .

For coordinate  $i$ , write  $p_{t,i}(v | a) = (1-t)q_{0,i}^P(v) + t\mathbf{1}\{v = a\}$ ,  $\dot{p}_{t,i}(v | a) = \mathbf{1}\{v = a\} - q_{0,i}^P(v)$  and  $S_{t,i}(a) = \{v : p_{t,i}(v | a) > 0\}$ . The implementation's minimum univariate conditional rate is, for  $u \neq v$ ,

$$R_{t,i}^*(u, v | a) = \frac{[\dot{p}_{t,i}(v | a) - \dot{p}_{t,i}(u | a)]_+ \mathbf{1}\{v \in S_{t,i}(a)\}}{|S_{t,i}(a)| p_{t,i}(u | a)}, \quad (57)$$

$$R_{t,i}^*(u, u | a) = - \sum_{v \neq u} R_{t,i}^*(u, v | a). \quad (58)$$

Full source support makes the off-diagonal denominator positive for the integration times used by the sampler. For  $x \neq y$ , define the local joint conditional generator

$$R_t^*(x, y \mid x_1, P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^*(x_i, y_i \mid x_{1,i}). \quad (59)$$

It permits at most one coordinate to jump at an infinitesimal time. The denoiser induces the ideal marginal coordinate rate

$$R_{t,i}^\theta(u, v \mid x, P) = \sum_{a \in \mathcal{X}_i} p_{\theta,i}(a \mid x, t, P) R_{t,i}^*(u, v \mid a), \quad (60)$$

and hence, off diagonal,

$$R_t^\theta(x, y \mid P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^\theta(x_i, y_i \mid x, P), \quad R_t^\theta(x, x \mid P) = - \sum_{y \neq x} R_t^\theta(x, y \mid P). \quad (61)$$

Fixed coordinates have identically zero off-diagonal rates.

For a grid  $0 = t_0 < \dots < t_{L_{\text{samp}}} = 1$  and  $\Delta t_k = t_{k+1} - t_k$ , an uncapped independent-coordinate Euler kernel would use

$$K_{k,i}^\theta(v \mid x) = \Delta t_k R_{t_k,i}^\theta(x_i, v \mid x, P), \quad v \neq x_i, \quad (62)$$

$$K_{k,i}^\theta(x_i \mid x) = 1 - \Delta t_k \sum_{v \neq x_i} R_{t_k,i}^\theta(x_i, v \mid x, P), \quad (63)$$

and  $\tilde{K}_k^\theta(y \mid x, P) = \mathbf{1}[y_{\mathcal{F}(P)} = x_{\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} K_{k,i}^\theta(y_i \mid x)$ . These probabilities are nonnegative when  $\Delta t_k \sum_{v \neq x_i} R_{t_k,i}^\theta(x_i, v \mid x, P) \leq 1$  for every coordinate and context.

The actual implementation uses one Monte Carlo clean endpoint  $A_{k,i} \sim p_{\theta,i}(\cdot \mid X_k, t_k, P)$  per coordinate. It substitutes  $R_{t_k,i}^*(\cdot, \cdot \mid A_{k,i})$  and multiplies every off-diagonal mass by

$$\alpha_{k,i} = \min \left\{ 1, \frac{0.999}{\Delta t_k \sum_{v \neq X_{k,i}} R_{t_k,i}^*(X_{k,i}, v \mid A_{k,i})} \right\}, \quad (64)$$

with  $\alpha_{k,i} = 1$  when the denominator is zero. The diagonal receives the remaining mass. Coordinates are then drawn independently,  $X_{k+1} \sim \prod_i \hat{K}_{k,i}$ , followed by  $\Pi_P$ . This cap makes every finite-step kernel nonnegative and stochastic. It also means the code uses a finite-step Monte Carlo Euler approximation rather than exact Gillespie simulation of the ideal CTMC.

**Proposition E.5** (Program-fixed invariance). *For every  $P \in \mathcal{P}_{\text{adm}}$ , corruption and numerical sampling satisfy  $\Pr[X_t \in \Omega_P] = 1$  and  $\Pr[X_k \in \Omega_P \text{ for all } k = 0, \dots, L_{\text{samp}}] = 1$ .*

*Proof.* The source and conditional path assign zero mass outside the fixed face. Fixed coordinates also have zero off-diagonal rates. Finally,  $\Pi_P(x) \in \Omega_P$  for every raw  $x$ , so the implementation safeguard preserves the claim by induction over finite steps.  $\square$

Let  $X_1 = \text{Enc}_P(G, \alpha)$  and define the population law

$$\nu(dG, d\alpha, dP, dt, dX_t) = \mu_{\text{bal}}(dG, d\alpha, dP) \mathcal{U}(dt) q_{t|1}^P(dX_t \mid X_1). \quad (65)$$

For  $i \in \mathcal{V}(P)$ , write  $\rho_i(a \mid X_t, t, P) = \Pr_\nu[X_{1,i} = a \mid X_t, t, P]$ .

**Proposition E.6** (Bayes optimality of the primary coordinatewise masked-denoising loss). *Assume an unrestricted coordinatewise conditional model class and positive state-family weight. On the event  $i \in \mathcal{V}(P)$ , every population minimizer of the primary masked*

denoising term satisfies

$$p_{\theta,i}^*(a \mid X_t, t, P) = \rho_i(a \mid X_t, t, P) \quad (66)$$

for every  $a \in \mathcal{X}_i$ ,  $\nu$ -almost surely. This proposition does not characterize the optimizer produced by auxiliary losses, finite capacity or PCGrad.

*Proof.* Fix a variable coordinate and context. Its conditional cross-entropy is  $H(\rho_i) + D_{\text{KL}}(\rho_i \parallel p_{\theta,i})$ , which is minimized exactly at  $p_{\theta,i} = \rho_i$ . The positive family weight and positive normalization by the number of variable coordinates do not alter that pointwise minimizer. Fixed coordinates do not appear in the primary loss.  $\square$

**Corollary E.7** (Population recovery of the local rate). *Suppose the conditional joint generator decomposes coordinatewise as above. Substituting any population minimizer  $p_{\theta,i}^*$  of the primary masked denoising term into the marginal-rate formula recovers the population local generator  $R_t$  almost surely on the training path.*

*Proof.* The proposition identifies  $p_{\theta,i}^*$  with the true clean-endpoint posterior marginal. The displayed rate is linear in that marginal, so substituting it performs the conditional expectation of  $R_{t,i}^*$  required for each local coordinate generator. Summing the one-coordinate terms recovers the joint local generator.  $\square$

**Formal path and proof of Theorem 2.1.** For the fixed program in the theorem, write the encoded endpoint mass function and population path as

$$q_1^P(x) = \Pr_{\mu_{\text{bal}}} [\text{Enc}_P(G, \alpha) = x \mid P], \quad (67)$$

$$q_t^P(x) = \sum_{x_1} q_1^P(x_1) q_{t|1}^P(x \mid x_1). \quad (68)$$

The theorem's exact-simulation assumption uses the displayed rates  $R_{t,i}^*$  on  $0 \leq t < 1$ , initial law  $X_0 \sim q_0^P$  and terminal law defined by  $t \uparrow 1$ . We prove the stronger path identity

$$\mathcal{L}(X_t \mid P) = q_t^P \quad (0 \leq t < 1), \quad \mathcal{L}(X_1 \mid P) = q_1^P. \quad (69)$$

For clarity, its conditional graph mass function is

$$\mu_G^P([G]_{\cong}) = \Pr_{\mu_{\text{bal}}} [[G']_{\cong} = [G]_{\cong} \mid P], \quad (70)$$

and the remaining conclusions are

$$\text{Dec}_{\#} q_1^P = \mu_G^P \quad (71)$$

on  $\mathfrak{G}_{N,K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}$ , with zero mass at  $\perp_{\text{dec}}$ , and

$$G_1 := \text{Dec}(X_1), \quad \Pr \left[ G_1 \in \mathfrak{G}_{N,K_{\text{cl}}}, \text{Valid}(G_1) = 1, \widehat{\text{Ll}}(G_1, P) = 1 \mid P \right] = 1. \quad (72)$$

*Proof of Theorem 2.1.* Fix an endpoint  $x_1$ . For one coordinate, let  $p_t(v) = p_{t,i}(v \mid x_{1,i})$  and  $\dot{p}_t(v) = \dot{p}_{t,i}(v \mid x_{1,i})$ . The off-diagonal definition of  $R_{t,i}^*$  gives the net probability flux into state  $v$  as

$$\sum_u p_t(u) R_{t,i}^*(u, v \mid x_{1,i}) = \frac{1}{|S_{t,i}|} \sum_{u \in S_{t,i}} ([\dot{p}_t(v) - \dot{p}_t(u)]_+ - [\dot{p}_t(u) - \dot{p}_t(v)]_+) \quad (73)$$

$$= \dot{p}_t(v) - \frac{1}{|S_{t,i}|} \sum_{u \in S_{t,i}} \dot{p}_t(u) = \dot{p}_t(v), \quad (74)$$

because the derivative of a probability vector sums to zero. Thus each univariate rate realizes its coordinate path. The sum of one-coordinate generators realizes the product conditional path  $q_{t|1}^P$  by the product rule:

$$\partial_t q_{t|1}^P(y | x_1) = \sum_x q_{t|1}^P(x | x_1) R_t^*(x, y | x_1, P). \quad (75)$$

Averaging this identity over  $q_1^P$  and applying Bayes' rule yields

$$\partial_t q_t^P(y) = \sum_x q_t^P(x) \sum_{x_1} \Pr[X_1 = x_1 | X_t = x, P] R_t^*(x, y | x_1, P) \quad (76)$$

$$= \sum_x q_t^P(x) R_t(x, y | P). \quad (77)$$

By Corollary E.7,  $R_t^{\theta^*} = R_t$  almost surely on the population path. On every compact interval  $[0, 1 - \eta]$ ,  $\eta > 0$ , this finite-state forward equation has a unique solution with initial law  $q_0^P$ , hence the exactly simulated CTMC has marginal  $q_t^P$ . Since each conditional coordinate path converges to its endpoint point mass,  $q_t^P \rightarrow q_1^P$  as  $t \uparrow 1$ .

Finally, deterministic decode round trip sends every encoded endpoint to its source graph. Summing  $q_1^P$  over all endpoint encodings of one graph class therefore gives  $\mu_G^P$ . The assumed endpoint validity and exact replay make the final probability-one statement immediate.  $\square$

**Corollary E.8** (Decoder contraction and approximate failure bound). *Under the endpoint assumptions of Theorem 2.1, let  $\tilde{q}_1^P$  be any approximate terminal raw-state law and define*

$$\mathcal{B}_P = \{x : \text{Dec}(x) = \perp_{\text{dec}}\} \cup \left\{x : \text{Dec}(x) = G \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(G) \widehat{\text{L1}}(G, P) = 0\right\}. \quad (78)$$

Then

$$\|\text{Dec}_{\#} \tilde{q}_1^P - \mu_G^P\|_{\text{TV}} \leq \|\tilde{q}_1^P - q_1^P\|_{\text{TV}}, \quad (79)$$

and

$$\Pr_{X \sim \tilde{q}_1^P}[X \in \mathcal{B}_P] \leq \|\tilde{q}_1^P - q_1^P\|_{\text{TV}}. \quad (80)$$

*Proof.* Total variation cannot increase under a measurable deterministic map, which proves the first inequality. The failure event has probability zero under  $q_1^P$  by Theorem 2.1; its probability under  $\tilde{q}_1^P$  is therefore at most the total-variation distance between the two raw-state laws.  $\square$

The theorem is intentionally an ideal population statement. It does not characterize the finite-capacity Transformer trained with auxiliary losses and PCGrad, does not bound the Monte Carlo endpoint approximation or capped Euler discretization, and does not imply support outside  $q_1^P$  or strong catalogue escape. Corollary E.8 identifies the single approximation quantity whose control would transfer ideal endpoint correctness to the implemented sampler.

## E.5 MATCHED-BUDGET POST-HOC FILTERING

Fix a reaction program  $P$ , and define

$$\mathcal{Y} = \mathfrak{G}_{N, K_{\text{cl}}} \cup \mathfrak{F}_{\text{gen}}, \quad A_P(y) = \mathbf{1}\left[y \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(y) = 1, \widehat{\text{L1}}(y, P) = 1\right]. \quad (81)$$

Thus  $A_P(y) = 0$  for invalid-decode and other generation failures. For a proposal law  $Q$  on  $\mathcal{Y}$ , conditional on a fixed trained model and requested program, let

$$Z_Q(P) = \Pr_{G \sim Q}[A_P(G) = 1] = \mathbb{E}_{G \sim Q}[A_P(G)] \quad (82)$$

denote the exact-L1 acceptance probability.

**Proposition E.9** (Exact law of matched-budget post-hoc filtering). *For  $B$  conditionally independent attempts  $Y_1, \dots, Y_B \sim Q$ ,  $N_B = \sum_{b=1}^B A_P(Y_b)$  satisfies*

$$N_B \sim \text{Binomial}(B, Z_Q(P)) \quad (83)$$

*for every  $Z_Q(P) \in [0, 1]$ . If  $Z_Q(P) > 0$ , rejection filtering additionally produces*

$$Q_{\text{post}}(g \mid P) = \Pr_{G \sim Q}[G = g \mid A_P(G) = 1] = \frac{Q(g)A_P(g)}{Z_Q(P)}. \quad (84)$$

*and consequently*

$$\mathbb{E}[N_B] = BZ_Q(P), \quad (85)$$

$$\text{Var}(N_B) = BZ_Q(P)(1 - Z_Q(P)), \quad (86)$$

$$\Pr[N_B = 0] = (1 - Z_Q(P))^B. \quad (87)$$

*If  $T_m$  is the number of independent attempts required to obtain  $m$  accepted products, then*

$$\mathbb{E}[T_m] = \frac{m}{Z_Q(P)}. \quad (88)$$

*Finally, for a program-conditioned proposal  $Q_P$  and an unconditioned proposal  $Q_0$ ,*

$$\mathbb{E}[N_B^P] - \mathbb{E}[N_B^0] = B(Z_{Q_P}(P) - Z_{Q_0}(P)). \quad (89)$$

*Proof.* Conditional independence makes the acceptance indicators Bernoulli with common probability  $Z_Q(P)$ , including the degenerate case  $Z_Q(P) = 0$ , so their sum is binomial. When  $Z_Q(P) > 0$ , Bayes’ rule gives

$$\Pr[G = g \mid A_P(G) = 1] = \frac{\Pr[A_P(G) = 1 \mid G = g] \Pr[G = g]}{\Pr[A_P(G) = 1]} \quad (90)$$

$$= \frac{A_P(g)Q(g)}{Z_Q(P)}. \quad (91)$$

The waiting time  $T_m$  has the negative-binomial distribution with success probability  $Z_Q(P)$ , whose expectation is  $m/Z_Q(P)$ . Applying the binomial expectation separately to  $Q_P$  and  $Q_0$  and subtracting proves the final identity.  $\square$

The proposition compares sampling efficiency only. It does not imply that conditioning improves  $Z_Q(P)$ ; the matched production experiment tests that empirical claim. The binomial count is the total accepted-attempt count, not the number of distinct accepted products.

## E.6 RECURSIVE ROUTE CONSTRUCTION

**Lemma E.10** (Termination of bounded route assessment). *Fix  $\mathfrak{B}_{\text{route}}$  with finite maximum route depth  $D_{\mathfrak{B}}$ . Assume finitely many candidate root decompositions, finitely many alternative expansions per molecule, finitely many reactants per expansion, and termination of every evidence, proposal and forward-verification call. Then  $\text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$  terminates in  $\mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}$ .*

*Proof.* We use induction on the remaining depth  $d = D_{\mathfrak{B}} - k$  at a node of depth  $k$ . When  $d = 0$ , the assessor performs the terminating evidence query and either accepts a terminal node, returns its typed failure, or rejects an expansion with a depth-exhaustion state. Suppose the claim holds whenever fewer than  $d > 0$  levels remain. At the current node, the finite sequence of terminating proposal, evidence and verification calls returns either a typed non-expansion outcome or one of finitely many finite reactant tuples. In the latter case, each child is assessed at depth  $k + 1$  and therefore has  $d - 1$  levels remaining. Every child assessment terminates by the induction hypothesis, and a finite tuple of finite child trees

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forms a finite parent tree. The finite root-decomposition loop also terminates. Rejecting an ancestor identity additionally prevents a circular molecular dependency from being accepted, but finite depth alone establishes termination.  $\square$

**Implementation invariant E.11** (Fail-closed synthesis-dossier implementation). *For the frozen assessor implementation, a successful return  $\text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) = T \in \mathfrak{D}$  is admitted only after checking  $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$ . Every other branch returns a member of the typed failure set  $\mathfrak{F}_{\mathfrak{B}}$ . This is an implementation invariant, not a completeness theorem for the chemical search space.*

The invariant records four independently checked conditions: the dossier is a well-formed tree rooted at  $G$ ; its selected root trace belongs to  $\widehat{\mathcal{E}}_P(G)$ ; every upstream reaction is evidence-admitted and satisfies  $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$ ; and every leaf closes under the dated availability policy. A typed failure means only that no complete dossier was returned under the frozen contract, not that no route exists under a larger search budget, a different admitted reaction set, or a later supplier snapshot.

## F GENERATED-STRUCTURE ATLAS

Table 13 reports 12 deterministic display examples from the same frozen seed-0 Ugi population used in Figure 2. The display begins with the lowest constitutional SHA-256 rank and then applies ECFP4 MaxMin selection among the 23 eligible, constitutionally unique products. This promotes visual coverage without using activity, route closure, procurement evidence or human preference. Each row includes the canonical constitutional SMILES so that the displayed identity is explicit; the exact strings are also stored in the hash-pinned atlas receipt.

Table 13: **FORGE generated-structure atlas**. The 12 rows form one multipage table, indexed by display order. Each constitutional graph is followed by its canonical SMILES. The decorated three-dimensional column shows one deterministic post-hoc RDKit ETKDGv3 structure optimized with up to 400 MMFF94s iterations. Purple denotes the amine-derived head, teal the aldehyde-derived body and tail, orange the isocyanide-derived tail and gray the assembly-introduced oxygen; these assignments are recovered through exact atom-mapped forward replay. FORGE does not generate three-dimensional coordinates, and these views are not predicted bioactive conformations. Exact L1 replay and descriptor-manifold membership are structural diagnostics, not evidence of activity, route closure or synthesis success. These are deterministic display examples, not experimental candidates.

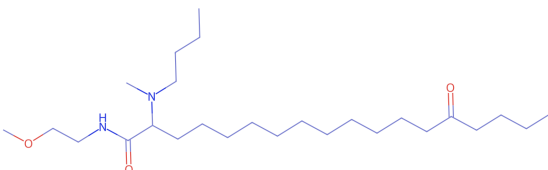
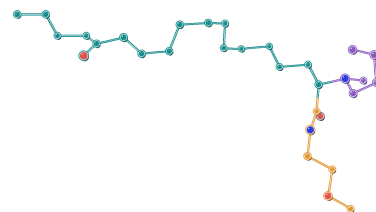
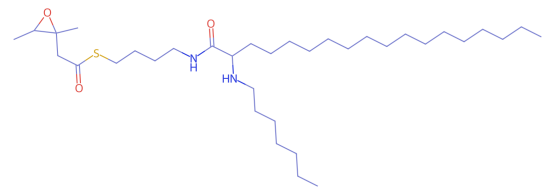
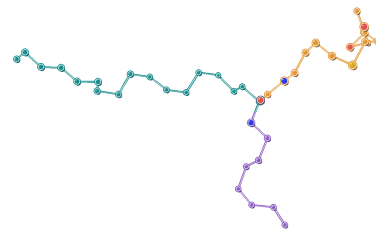
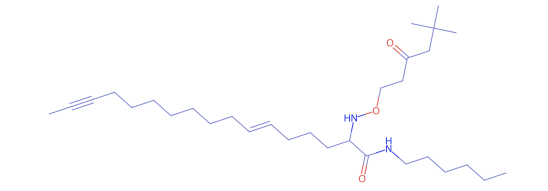
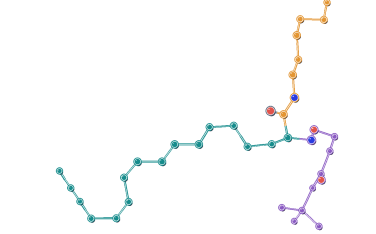
| CONSTITUTIONAL GRAPH AND CANONICAL SMILES |                                                                                                                                                          | DECORATED 3D VIEW                                                                    |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1                                         | <br><chem>CCCCC(=O)CCCCCCCCCCCC(C(=O)NCCCOC)N(C)CCCC</chem>            |   |
| 2                                         | <br><chem>CCCCCCCCCCCCCCCC(NCCCCCCC)C(=O)NCCCCSC(=O)CC1(C)OC1C</chem> |  |
| 3                                         | <br><chem>CC#CCCCCCCCC=CCCC(NCCCC(=O)CC(C)(C)C)C(=O)NCCCCC</chem>     |  |

Table 13, continued

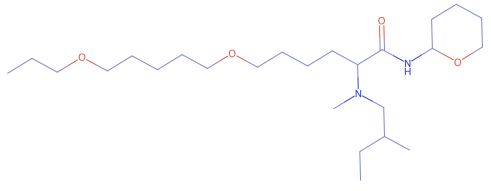
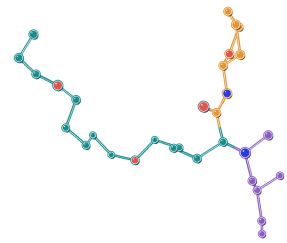
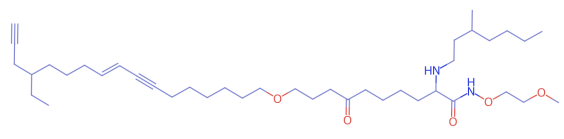
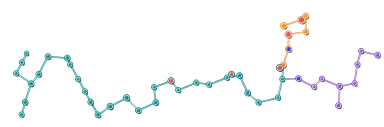
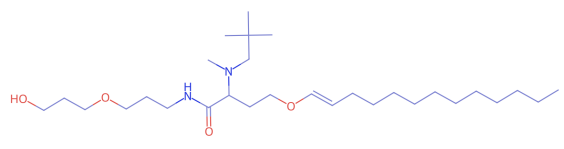
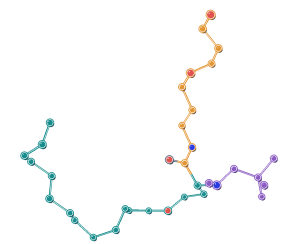
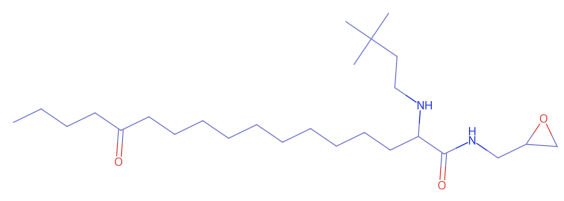
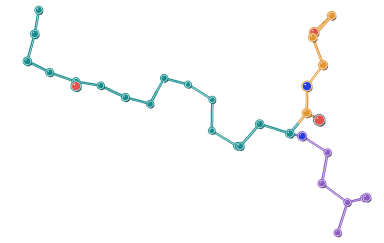
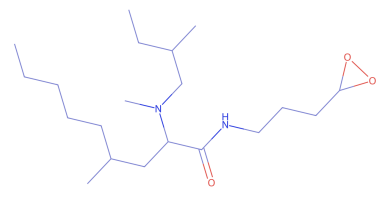
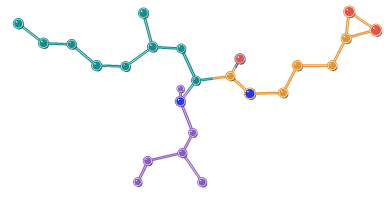
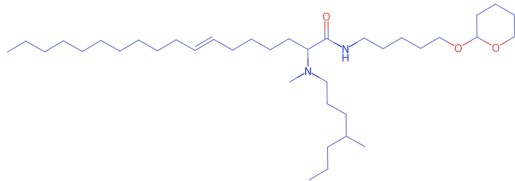
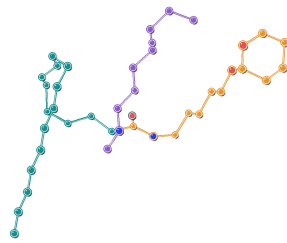
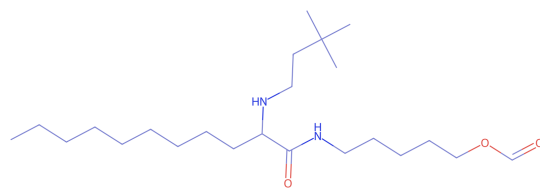
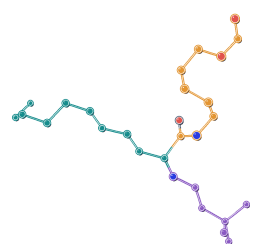
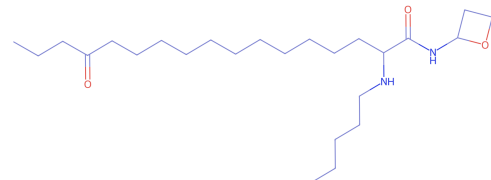
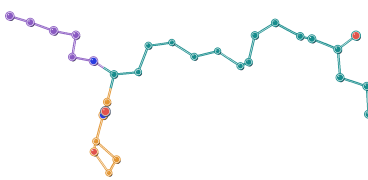
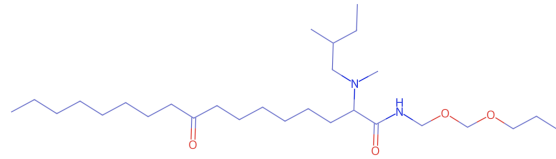
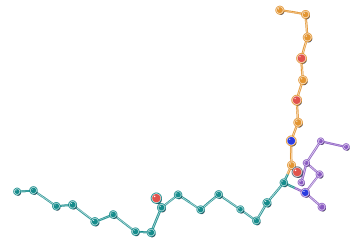
|   | CONSTITUTIONAL GRAPH AND CANONICAL SMILES                                                                                                                       | DECORATED 3D VIEW                                                                    |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 4 | <br><chem>CCCCCCCCCCCCC(=O)NC1CCCC1)N(C)CC(C)CC</chem>                         |    |
| 5 | <br><chem>C#CCC(CC)CCCC=CC#CCCCCCCCCCCC(=O)CCCC(NCCC(C)CCCC)C(=O)NCCCOC</chem> |    |
| 6 | <br><chem>CCCCCCCCCCCC=CCCC(C(=O)NCCC1CCCC1)N(C)CC(C)(C)C</chem>              |   |
| 7 | <br><chem>CCCCC(=O)CCCCCCCCCCCC(NCCC(C)(C)C)C(=O)NCC1C01</chem>              |  |
| 8 | <br><chem>CCCCC(C)CC(C(=O)NCCC1001)N(C)CC(C)CC</chem>                        |  |

Table 13, continued

|    | CONSTITUTIONAL GRAPH AND CANONICAL SMILES                                                                                                          | DECORATED 3D VIEW                                                                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 9  | <br><chem>CCCCCCCCC=CCCCC(C(=O)NCCCCOC1CCOC1)N(C)CCCC(C)CC</chem> |    |
| 10 | <br><chem>CCCCCCCCC(NCCC(C)(C)C)C(=O)NCCCCOC=O</chem>             |   |
| 11 | <br><chem>CCCCNC(CCCCCCCCCC(=O)CCC)C(=O)NC1CCO1</chem>           |   |
| 12 | <br><chem>CCCCCCCC(=O)CCCCC(C(=O)NCOCOC)N(C)CC(C)CC</chem>      |  |